LogicSplat: Neuro-Symbolic 3D Scene Graph Generation from Gaussian Splats via Geometric Kolmogorov-Arnold Networks

PREFACE

Understanding how objects relate to one another in three-dimensional space is among the more quietly difficult problems in computer vision. It is not enough for a system to detect that a table and a cup are present in a scene — it must understand that the cup sits *on* the table, that the router is *behind* the laptop, that the book is *lower than* the shelf it fell from. These spatial relations are the connective tissue of scene understanding, and yet they remain surprisingly hard to learn from visual data alone.

This project began with a broad interest in that problem and a specific context: 3D Gaussian Splatting, which has made high-quality scene reconstruction accessible from a few minutes of casual smartphone video. The original conception was ambitious — semantic segmentation via foundation models, natural language querying via a local LLM. What the research ultimately produced was something quite different, and I believe more interesting: a geometry-native system, free of any foundation model, that learns spatial relations directly from the geometric properties of Gaussian clusters. That shift was not a retreat from the original ambition. It was where the evidence led.

The report documents both the system that was built and the reasoning behind every design decision — the feature engineering, the training strategy, the architectural choices, and the evaluation across three distinct scene domains. The end-to-end user-facing application originally planned is not within the scope of this report; that remains natural future work, as discussed in the concluding chapter.

I am grateful to Mr. Sahil Shah, whose guidance throughout this project was decisive — particularly his suggestion to explore Geometric Kolmogorov-Arnold Networks at a point when the research needed direction. I also thank the open-source communities behind NerfStudio, PyTorch Geometric, and the 3RScan dataset for making this work possible.

Debjyoti Sengupta

Symbiosis Institute of Geoinformatics,

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1.INTRODUCTION

1.1 Background and Motivation

The ability to reason about the spatial relationships between objects in a three-dimensional environment is a fundamental requirement for machines that must interpret, navigate, or interact with the physical world. Whether the application is robotic manipulation, autonomous navigation, or augmented reality, a system cannot act meaningfully on a scene it can only perceive as a collection of isolated objects. It must also understand that one object rests on top of another, that two objects are adjacent, that a third is directly in front of a fourth. This structured, relational understanding of a scene is formally encoded as a scene graph — a directed graph in which nodes represent object instances and edges represent the spatial or semantic relationships between them.

Constructing such graphs from real three-dimensional data has historically required either expensive depth sensors, manually annotated point clouds, or large foundation models trained on internet-scale data. The emergence of 3D Gaussian Splatting (3DGS), introduced by Kerbl et al. (2023), offers a compelling alternative starting point. Unlike neural radiance fields, a Gaussian splat reconstructs a scene as an explicit set of anisotropic Gaussian primitives, each carrying position, colour, opacity, and a full 3D covariance matrix that encodes local surface geometry. Critically, a photorealistic Gaussian splat of a room-scale scene can now be reconstructed from a short smartphone video in under fifteen minutes using NerfStudio's splatfacto pipeline, placing high-quality 3D scene reconstruction within reach of end users without specialised hardware.

Despite this accessibility, the problem of extracting structured relational scene graphs directly from Gaussian splats remains largely unsolved. Existing approaches either rely on 2D image features from large vision-language models — introducing dependency on foundation models with billions of parameters and minute-scale inference times — or operate on mesh and point cloud representations that do not exploit the unique geometric richness of the Gaussian representation. The natural question arises: can the geometry encoded within the Gaussian primitives themselves — their positions, scales, and covariance structures — provide sufficient signal to predict spatial relations accurately, without any foundation model at inference time?

This project, LogicSplat, investigates that question. It proposes a geometry-first pipeline that proceeds from raw smartphone video, through Gaussian splat reconstruction and object clustering, to a scene graph predicted entirely from geometric features by a Geometric Kolmogorov-Arnold Network (GeoKAN). A zero-parameter symbolic consistency repair module enforces logical constraints on the output, ensuring the final graph is free of contradictions. The result is a system that achieves Macro F1 = 0.926 and R@5 = 98.3% on the 3DSSG benchmark using only 1.018 million parameters and under two seconds of inference time on a consumer GPU — outperforming foundation-model-based methods that are three orders of magnitude larger.

1.2 Problem Statement

Despite the rapid advancement of 3D scene reconstruction techniques, the automatic generation of structured spatial scene graphs from Gaussian splat representations remains an open and largely unaddressed problem. Three specific gaps define the current state of the field.

First, existing scene graph methods that operate on 3D data depend heavily on large foundation models — vision-language models such as CLIP, SAM, LLaVA, and GroundingDINO — to interpret object identity and spatial context. Systems such as GaussianGraph (Li et al., 2025) and ReLaGS (Arafa et al., 2026) require billions of parameters, several gigabytes of GPU memory, and over ten minutes of inference time per scene. This dependency makes them impractical for deployment on consumer hardware and unsuitable for real-time applications.

Second, the 3D Gaussian Splat representation carries rich geometric information in the form of per-Gaussian covariance matrices, opacity values, and spatial distributions — yet existing pipelines discard this geometry in favour of projected 2D image features. The question of whether the geometry encoded within Gaussian primitives is itself sufficient to predict spatial relations — without any visual language model — has not been systematically investigated.

Third, the annotations provided by the 3DSSG dataset (Wald et al., 2020), which is the primary benchmark for indoor 3D scene graphs, are extremely sparse: human annotators label only 80–120 directed object pairs per scene out of approximately 870 possible pairs. This severe annotation sparsity means that standard supervised training on 3DSSG labels leaves the majority of geometrically derivable relations unlearned, suppressing model performance independently of architectural choices.

This project addresses all three gaps. It asks whether a lightweight model trained exclusively on geometric features extracted from Gaussian splats, augmented with rule-derived labels to overcome annotation sparsity, and refined by a zero-parameter symbolic consistency module, can match or exceed the performance of foundation-model-dependent systems on established benchmarks — while remaining deployable on a single consumer GPU in under two seconds.

1.3 Research Questions

This project is structured around the following research questions:

RQ1. Can geometric features extracted directly from 3D Gaussian Splat primitives — without any visual language model or 2D image feature — provide sufficient signal to predict spatial relations between objects in indoor scenes?

RQ2. Does a Kolmogorov-Arnold Network architecture with a learnable geometric metric outperform standard graph neural network layers for spatial relation prediction, and does the choice of basis function (RBF, wavelet, or fixed diagonal) materially affect performance?

RQ3. To what extent can rule-based pseudo-label injection address the annotation sparsity inherent in the 3DSSG dataset, and how much of the observed performance gain is attributable to this injection rather than architectural improvements?

RQ4. Does a zero-parameter symbolic consistency repair module, enforcing logical constraints such as inverse completeness, mutual exclusion, and transitivity, improve scene graph quality on held-out and cross-domain scenes without retraining?

RQ5. How well does a model trained exclusively on room-scale indoor scenes (3RScan, 3–8 m scene extent) generalise zero-shot to a significantly different scale domain — specifically tabletop environments with scene extents of 0.3–0.8 m — and which spatial relations transfer reliably across this domain gap?

RQ6. Does the geometry-first pipeline generalise to indoor scenes from an entirely different dataset (LERF), and is the degree of generalisation explained by scale similarity rather than scene type?

RQ7. How does the computational footprint of the proposed geometry-first approach compare against foundation-model-dependent systems in terms of parameter count, GPU memory, and inference time, at equivalent or superior benchmark performance?

RQ8. Can the complete pipeline — from consumer smartphone video through Gaussian splat reconstruction to final scene graph — be executed end-to-end on a single consumer GPU within a practically deployable time budget?

1.4 Research Objectives

The overarching objective of this project is to design, implement, and evaluate a lightweight, geometry-first pipeline for spatial scene graph generation from 3D Gaussian Splats, operating without foundation models at inference time. The specific objectives are as follows:

O1. To develop a complete end-to-end pipeline that converts a short smartphone video into a structured spatial scene graph, encompassing Gaussian splat reconstruction, object clustering, geometric feature extraction, relation prediction, and symbolic consistency repair.

O2. To design and implement the GeoKAN layer — a Kolmogorov-Arnold Network variant with a learnable geometric metric — and integrate it into a dual-head graph neural network architecture (GeoKANRelationGNN) capable of jointly predicting contact and directional spatial relations.

O3. To develop a rule-based label injection strategy that derives geometrically certain pseudo-labels for unannotated object pairs in the 3DSSG dataset, thereby substantially increasing the effective training signal without introducing data leakage.

O4. To design and implement a zero-parameter symbolic consistency repair module (SceneGraphRepair) that enforces inverse completeness, mutual exclusion, asymmetry, and transitivity constraints on predicted scene graphs through fixed-point iteration.

O5. To conduct a systematic ablation study comparing three GeoKAN basis function variants — RBF, Mexican hat wavelet, and fixed diagonal (Gamma) — under identical training conditions, and to identify the variant that achieves the best balance of accuracy and training stability.

O6. To benchmark the final system against state-of-the-art scene graph methods on the 3DSSG validation set and the LERF dataset, and to evaluate zero-shot cross-domain generalisation on eight custom-captured tabletop scenes.

O7. To characterise the domain gap between room-scale training data and tabletop evaluation data, identifying which spatial relations transfer robustly and which degrade, and explaining the root cause in terms of scale-dependent feature distributions.

1.5 Scope and Delimitations

This project focuses on the prediction of spatial relations between object instances in static three-dimensional indoor scenes reconstructed as 3D Gaussian Splats. The scope and boundaries of the work are defined as follows.

Within scope:

The project covers the full pipeline from Gaussian splat input to scene graph output, including opacity-based filtering, statistical outlier removal, RANSAC-based background segmentation, HDBSCAN object clustering, geometric feature engineering, graph neural network inference, per-relation threshold calibration, and symbolic consistency repair. Evaluation is conducted on the 3DSSG benchmark (Wald et al., 2020), four LERF scenes (Kerr et al., 2023), and eight custom-captured tabletop scenes. The relation schema consists of ten directed spatial relations: on_top_of, under, attached_to, adjacent_to, left_of, right_of, in_front_of, behind, higher_than, and lower_than.

Outside scope:

The project does not address semantic object recognition or open-vocabulary object labelling as a primary contribution; object identity labels are used only for human-readable output and do not influence relation prediction. Dynamic scenes, scenes with significant occlusion, and outdoor environments are not considered. The project does not address scene graph generation from raw RGB-D sequences, mesh-based representations, or NeRF implicit fields. Temporal reasoning — tracking how spatial relations change over time — is outside the scope of this work.

The project does not claim to produce a deployable robotics system or a generalised spatial reasoning engine. The tabletop evaluation scenes were captured and annotated by the author specifically for cross-domain testing; they do not constitute a large-scale benchmark and results on them are reported as indicative of zero-shot generalisation rather than definitive cross-domain performance.

Finally, the two relations — INSIDE and HANGING_FROM — present in the original 3DSSG vocabulary are excluded from the relation schema on account of insufficient positive training examples (262 and 974 instances respectively out of over two million total edges), and their prediction is not attempted.

1.6 Significance of the Study

The significance of this work lies at the intersection of three practical concerns: accessibility of 3D scene understanding, efficiency of spatial reasoning systems, and the integrity of evaluation methodology in cross-domain machine learning.

From a technical standpoint, this project demonstrates that foundation models are not a prerequisite for high-quality spatial scene graph generation. The proposed system achieves Macro F1 = 0.926 and R@5 = 98.3% on the 3DSSG validation benchmark using 1.018 million parameters and under two seconds of inference time on a consumer GPU — outperforming ReLaGS (Arafa et al., 2026), which uses approximately 1.5 billion parameters and requires 12.6 minutes per scene. This result has direct implications for the deployability of 3D scene understanding in resource-constrained environments, including edge devices, mobile robotics platforms, and applications where low latency is essential.

From a representational standpoint, the project makes a concrete contribution to the understanding of what geometric information Gaussian Splats contain and how it can be exploited. By demonstrating that 10-dimensional node features and 22-dimensional edge features derived entirely from cluster geometry — without colour, texture, or semantic labels — are sufficient to learn ten spatial relation classes to high accuracy, the project establishes a principled baseline for geometry-first scene understanding from Gaussian representations.

From a methodological standpoint, the project contributes an honest account of cross-domain evaluation under a genuine domain gap. The 11-percentage-point difference between an inflated R@3 obtained through post-hoc ground truth adjustment (0.864) and the honest zero-shot result (0.758) is documented and explained, providing the community with a clear illustration of how evaluation design choices can misrepresent generalisation. The final reported numbers — R@3 = 0.758, R@5 = 0.927 on tabletop scenes — are derived from ground truth constructed purely from deterministic geometric principles, with no adaptation to model failures.

Finally, the symbolic consistency repair module contributes a domain-agnostic, zero-parameter post-processing strategy that improves Micro F1 by 3.4 points on cross-domain data solely through logical inference, without any retraining. This establishes a reusable component that can be applied to any scene graph system whose output is expected to satisfy spatial logic constraints.

1.7 Structure of the Report

The remainder of this report is organised as follows.

Chapter 2: Literature Review surveys the key prior work across six areas relevant to this project: 3D scene understanding and scene graphs, 3D Gaussian Splatting, graph neural networks for relational reasoning, Kolmogorov-Arnold Networks, neuro-symbolic approaches in 3D vision, and symbolic consistency in scene understanding. The chapter concludes with an identification of the research gap that motivates this work.

Chapter 3: Methodology presents the complete technical pipeline. It covers the scene reconstruction process from smartphone video through COLMAP and NerfStudio splatfacto, Gaussian data preprocessing, HDBSCAN-based object clustering, geometric feature engineering, rule-based label injection, the GeoKANRelationGNN architecture and its three basis function variants, the training strategy including asymmetric loss and data augmentation, symbolic consistency repair, and the evaluation protocol for all three test settings.

Chapter 4: Results reports the quantitative outcomes of all experiments. It presents the model evolution table from the GAT baseline through GeoKAN-Gamma, the basis function ablation, comparison against state-of-the-art methods on the 3DSSG benchmark, per-scene results on the LERF held-out evaluation, the impact of symbolic repair across datasets, zero-shot cross-domain results on eight tabletop scenes, and a computational efficiency comparison.

Chapter 5: Discussion interprets the results in depth, examining why geometry alone proves sufficient, why the fixed diagonal GeoKAN-Gamma variant outperforms the adaptive RBF variant, how rule-based label injection drives the majority of the performance improvement, and what the cross-domain results reveal about the nature of the scale domain gap. Limitations of the current work are also addressed.

Chapter 6: Conclusion summarises the contributions of the project, states the novel findings, and outlines directions for future work.

2.LITERATURE REVIEW

2.1 3D Reconstruction and Representation

Kerbl et al. (2023) — 3D Gaussian Splatting

Kerbl et al. introduced 3D Gaussian Splatting (3DGS) as an explicit, rasterisation-based alternative to implicit neural radiance fields for real-time novel view synthesis. Rather than encoding a scene as a continuous neural function, 3DGS represents it as a set of anisotropic three-dimensional Gaussian primitives, each parameterised by a position, an opacity value, a colour encoded as spherical harmonic coefficients, a scale vector, and a rotation quaternion. Together, the scale and rotation define a full 3×3 covariance matrix that describes the local geometric shape of each Gaussian. This explicit representation renders at real-time frame rates and supports direct access to per-Gaussian geometry — a property that distinguishes 3DGS from all prior radiance field approaches. For this project, the Gaussian splat is not merely a rendering artefact but the primary data structure from which object instances are clustered and spatial relations are inferred. The covariance matrices, opacity values, and positions of individual Gaussians constitute the

raw geometric signal from which all 10-dimensional node features and 22-dimensional edge features are derived.

Tancik et al. (2023) — NerfStudio

Tancik et al. introduced NerfStudio as a modular, open-source framework that unified the training, evaluation, and deployment of neural radiance field methods under a common interface. NerfStudio includes splatfacto, an implementation of the 3D Gaussian Splatting training pipeline adapted for practical use from consumer-captured video. The framework handles COLMAP-based camera pose estimation, Gaussian initialisation from sparse point clouds, and iterative densification and pruning over 30,000 training iterations. In this project, NerfStudio splatfacto serves as the reconstruction engine that converts a short smartphone video — typically 60 frames at 640×480 resolution — into the .ply Gaussian splat file that enters the LogicSplat pipeline. The accessibility of NerfStudio is directly relevant to the practical motivation of the project: it makes high-quality 3D reconstruction available without specialised hardware.

Kerr et al. (2023) — LERF

Kerr et al. proposed Language Embedded Radiance Fields (LERF), a method that distils CLIP language features into a NeRF volume to enable open-vocabulary localisation within a scene by querying arbitrary text phrases. LERF released four publicly available scenes — ramen, teatime, waldo_kitchen, and figurines — each reconstructed from over 100 frames and covering indoor environments at 1–5 metre scale. While LERF's primary contribution is language grounding rather than scene graph generation, its public scene releases have been adopted as a standard evaluation environment for 3D scene understanding methods, including GaussianGraph (Li et al., 2025). In this project, the four LERF scenes serve as a held-out evaluation set for testing in-domain generalisation to an unseen dataset. The scenes are not used for training or threshold calibration, and ground truth spatial relations are derived geometrically from cluster centroids rather than from LERF's language features.

2.2 Three-Dimensional Scene Graph Generation

Wald et al. (2020) — 3DSSG

Wald et al. introduced the 3D Semantic Scene Graph (3DSSG) dataset and established the task of learning scene graph prediction from 3D indoor reconstructions. Built on top of the 3RScan scan sequences, 3DSSG provides human-annotated pairwise spatial and semantic relationships between object instances across 478 scenes, with a relation vocabulary of 26 types covering spatial, comparative, and semantic predicates. The annotation process relied on crowdsourced human labellers who identified visible and physically plausible relationships between object pairs within each scene. A fundamental limitation acknowledged in the dataset is annotation sparsity: labellers annotated only 80–120 directed object pairs per scene out of approximately 870 possible directed pairs in a typical scene, leaving approximately 90 per cent of geometrically valid pairs unannotated. This sparsity is a central challenge addressed in this project through rule-based label injection, which derives pseudo-labels for unannotated directional and contact pairs, increasing the effective training signal by 8–11 times. The 3DSSG validation set of 85 held-out scenes, containing 323,237 ground truth relation triples, serves as the primary in-domain benchmark for this project.

Wu et al. (2021) — SceneGraphFusion

Wu et al. proposed SceneGraphFusion, an incremental 3D scene graph prediction system that constructs and updates a scene graph in real time from an RGB-D video stream. The method maintains a factor graph over object instances, with node and edge features updated as new depth frames are integrated. SceneGraphFusion demonstrated that scene graph generation from 3D data is feasible in an online setting without requiring a complete reconstruction as a prerequisite, achieving competitive results on the 3DSSG benchmark. However, its dependence on real-time depth sensor input and frame-by-frame processing makes it structurally incompatible with the post-hoc, reconstruction-based setting of this project. SceneGraphFusion's contribution is significant in establishing that graph neural networks can operate on 3D geometric features for relation prediction, a core insight that this project extends by operating on Gaussian splat geometry rather than RGB-D depth maps.

Koch et al. (2024) — Open3DSG

Koch et al. introduced Open3DSG, the first method to predict open-vocabulary 3D scene graphs from point cloud inputs, enabling the prediction of object classes and relation types not present in the training label set. The approach co-embeds 3D scene graph features with the feature space of CLIP-based vision-language models, allowing arbitrary semantic queries at inference time. On the 3DSSG benchmark, Open3DSG reports R@3 = 0.580 and R@5 = 0.650. While the open-vocabulary capability represents a meaningful advance over closed-set methods, it introduces a dependency on CLIP (approximately 400 million parameters) at inference time. This project surpasses Open3DSG on both R@3 (+29.8 percentage points) and R@5 (+33.3 percentage points) while using a model that is over 390 times smaller and requires no vision-language model at inference.

Gu et al. (2024) — ConceptGraphs

Gu et al. introduced ConceptGraphs, a system that constructs open-vocabulary 3D scene graphs from sequences of posed RGB-D images by fusing outputs from 2D foundation models — including class-agnostic instance segmentation, CLIP feature extraction, and large language model relation inference — into a persistent 3D map. ConceptGraphs produces human-interpretable scene graphs suitable for downstream robotic planning and natural language querying. On the 3DSSG benchmark it achieves R@3 = 0.740 and R@5 = 0.790. The system's reliance on an LLM for relation inference introduces both significant computational overhead and non-determinism at inference time. The present project achieves R@3 = 0.878 and R@5 = 0.983 — exceeding ConceptGraphs by 13.8 and

19.3 percentage points respectively — using a deterministic, geometry-only model with no language model dependency.

2.3 Scene Understanding from Gaussian Splats

Qin et al. (2024) — LangSplat

Qin et al. introduced LangSplat, a method that augments 3D Gaussian Splats with per-Gaussian CLIP language features to enable open-vocabulary localisation and semantic querying within a scene. Rather than querying a neural volume as in LERF, LangSplat trains a small autoencoder to compress CLIP features and distils them directly into the Gaussian primitives, achieving significantly faster query speeds. LangSplat demonstrated that the Gaussian representation is amenable to feature embedding beyond colour and geometry, and established a pattern — language-augmented Gaussians — that several subsequent works have followed. However, LangSplat's semantic understanding is pixel-level rather than instance-level: it localises language queries to regions of Gaussians rather than to discrete object instances with explicit boundaries. This limitation motivates the object-centric approach of this project, in which Gaussians are first segmented into discrete object clusters before any relational reasoning is performed.

Ye et al. (2024) — Gaussian Grouping

Ye et al. proposed Gaussian Grouping, a method that extends 3DGS with per-Gaussian identity features trained using 2D segmentation masks from the Segment Anything Model (SAM), enabling instance-level segmentation and editing of 3D Gaussian scenes. By assigning each Gaussian a learned group identity embedding, Gaussian Grouping can segment, remove, inpaint, or recolour individual object instances within a splat. This work is directly relevant to the object clustering problem that this project addresses: Gaussian Grouping solves instance segmentation using SAM supervision, whilst this project solves it using geometry-only HDBSCAN clustering with no external segmentation model. The comparison is instructive — Gaussian Grouping achieves sharper instance boundaries at the cost of requiring SAM at reconstruction time, whereas the geometry-only approach of this project operates post-hoc on any existing splat without retraining.

Li et al. (2025) — GaussianGraph

Li et al. proposed GaussianGraph, a framework for open-world scene graph generation from 3D Gaussian Splats that integrates adaptive semantic clustering with a multi-modal relation prediction pipeline. GaussianGraph introduces a Control-Follow clustering strategy that dynamically adapts to scene scale and avoids CLIP feature compression, followed by a relation prediction stage that combines geometric verification with vision-language model outputs from CLIP, SAM, LLaVA-1.6, and GroundingDINO. On the four LERF evaluation scenes, GaussianGraph with its 3D correction module achieves mR@5 = 63.2% using positional relation queries. This project achieves mR@5 = 92.9% on the

same scenes — a 29.7 percentage point improvement — using a model with no foundation model dependency at inference. GaussianGraph serves as the primary baseline for the LERF evaluation in this project, and its 3D correction module is qualitatively compared against the symbolic consistency repair approach proposed here.

Arafa et al. (2026) — ReLaGS

Arafa et al. introduced ReLaGS (Relational Language Gaussian Splatting), a framework that constructs hierarchical language-distilled Gaussian scene representations and their corresponding 3D semantic scene graphs without scene-specific training. ReLaGS employs a Gaussian pruning mechanism to refine scene geometry, a multi-view language alignment strategy to aggregate 2D vision-language features into accurate 3D object embeddings, and a graph neural network to predict spatial and semantic relations from the resulting embeddings. Accepted at CVPR 2026, ReLaGS represents the current state of the art among Gaussian-based scene graph methods, reporting R@3 = 0.790 and R@5 = 0.870 on the 3DSSG benchmark using approximately 1.5 billion parameters and 12.6 minutes of inference time per scene on 7.5 GB of GPU memory. This project surpasses ReLaGS on both metrics — R@3 = 0.878 and R@5 = 0.983 — using 1.018 million parameters, under two seconds of inference, and less than 1.5 GB of GPU memory, with no foundation model at inference time.

2.4 Datasets

Wald et al. (2019) — 3RScan

Wald et al. introduced 3RScan, a large-scale dataset of indoor RGB-D scan sequences designed for the task of 3D object instance re-localisation across scene changes. The dataset contains 1,482 scenes captured across multiple temporal sessions using a Structure Sensor depth camera, with each scene reconstructed as a textured mesh annotated with per-vertex instance labels. The re-scan design — where the same room is captured at different points in time with objects moved or replaced — makes 3RScan unique among indoor datasets in explicitly modelling scene dynamics. For this project, 3RScan serves as the primary source of training and validation data. Gaussian splat reconstructions of 565 3RScan scenes are sourced from the GaussianWorld repository, which provides MCMC-based 3DGS reconstructions of the original scan sequences. Of these, 480 scenes are used for training and 85 for validation, with instance labels transferred from the original mesh vertex annotations to Gaussian splat centres via nearest-neighbour assignment, achieving a mean coverage of 82.7 per cent.

Dai et al. (2017) — ScanNet

Dai et al. introduced ScanNet, a richly annotated dataset of 1,513 indoor RGB-D scans covering 707 distinct spaces, providing per-frame depth images, camera poses, surface reconstructions, and per-vertex semantic and instance segmentation labels. ScanNet became one of the most widely used benchmarks for indoor 3D scene understanding, providing standardised splits and evaluation protocols for semantic segmentation, object detection, and scene reconstruction tasks. In this project, ScanNet was used during an

exploratory phase of model development — specifically for training early GAT-based baseline models (v1 through v7) using geometry-derived pseudo-labels prior to the transition to 3RScan. Although the ScanNet-trained models achieved strong in-domain performance, they failed to generalise to custom tabletop scenes due to the cross-domain gap between ScanNet's room-scale reconstructions and the tabletop evaluation environment. This finding motivated the move to 3RScan with human-annotated 3DSSG labels as the primary training source.

2.5 Graph Neural Networks

Veličković et al. (2018) — Graph Attention Networks

Veličković et al. introduced Graph Attention Networks (GAT), a graph neural network architecture that computes node representations by attending over each node's neighbourhood using a learnable attention mechanism. Unlike prior spectral graph convolution methods, GAT operates in the spatial domain and is inductive — it can generalise to unseen graph structures without retraining. The attention coefficients are computed as a function of the features of the source and target nodes, allowing the network to assign different importance to different neighbours during message passing. GAT became one of the most widely adopted graph neural network architectures for tasks involving irregular, non-Euclidean data, including molecular property prediction, citation network classification, and 3D scene understanding. In the context of this project, GAT-based models formed the baseline architecture during the early ScanNet training phase, demonstrating that graph-structured relational reasoning over 3D object features was a viable approach before the transition to the GeoKAN architecture.

Brody et al. (2021) — GATv2

Brody et al. identified a fundamental expressiveness limitation in the original GAT: the attention function factorises in a way that produces a static ranking of neighbours independent of the query node, meaning the model cannot compute dynamic attention in the general case. GATv2 addresses this by modifying the order of operations — applying the linear transformation after concatenating source and target node features, rather than before — which restores the full expressive power of dynamic attention. On multiple benchmarks, GATv2 demonstrates measurably improved performance over GAT on tasks where dynamic neighbourhood weighting is important. In this project, two GATv2Conv layers with four attention heads and a 22-dimensional edge feature input form the message passing backbone of GeoKANRelationGNN. The edge features — encoding directional offsets, volume ratios, and contact scores between object pairs — are incorporated directly into the attention computation, allowing the graph attention mechanism to weight neighbouring objects based on their geometric relationship to the query object.

2.6 Kolmogorov-Arnold Networks

Liu et al. (2024) — KAN

Liu et al. introduced Kolmogorov-Arnold Networks (KANs) as a learnable alternative to the multi-layer perceptron, motivated by the Kolmogorov-Arnold representation theorem — a classical result in approximation theory stating that any continuous multivariate function can be expressed as a finite composition of continuous univariate functions and addition. Whereas a standard MLP places fixed activation functions on nodes and learnable weights on edges, a KAN places learnable activation functions on edges, parameterised as B-splines, with no fixed nonlinearity. This design allows each edge to learn an arbitrary univariate transformation, giving the network the capacity to represent complex functional relationships more efficiently than an MLP of equivalent parameter count. Liu et al. demonstrated that KANs outperform MLPs on scientific function fitting and partial differential equation solving tasks, often with dramatically fewer parameters, and exhibit greater interpretability by virtue of their univariate edge functions being directly visualisable. This work provides the theoretical and architectural foundation for the GeoKAN layer employed in this project.

Sen et al. (2026) — GeoKAN

Sen et al. introduced Geometric Kolmogorov-Arnold Networks (GeoKAN), a family of geometry-aware KAN-type models in which basis expansion is performed in learned, geometry-adapted coordinates rather than in the fixed Euclidean input space. GeoKAN achieves this by learning a diagonal Riemannian metric that warps the input before basis expansion — the metric scales each input dimension by a learned factor, effectively stretching or compressing the feature space to align with the geometry of the data manifold. Three main variants are developed: GeoKAN-γ, which uses a fixed diagonal metric parameterised by a learnable scalar per dimension; GeoKAN-NNMetric, which computes an input-adaptive metric via a small neural network; and LM-KAN, which extends both variants with RBF, wavelet, and Fourier basis functions. Sen et al. demonstrate improved approximation of complex targets in supervised fitting and physics-informed learning compared to standard KANs and MLPs. This project directly applies the GeoKAN-γ and LM-KAN-RBF variants to the spatial relation prediction task, implementing them as the classification heads of GeoKANRelationGNN and demonstrating for the first time that GeoKAN's geometry-adaptive metric provides measurable benefit for structured prediction over 3D scene graphs.

2.7 Training Strategy for Sparse and Imbalanced Labels

Ridnik et al. (2021) — Asymmetric Loss

Ridnik et al. introduced the Asymmetric Loss (ASL) for multi-label image classification, motivated by the severe positive-negative imbalance characteristic of multi-label datasets where most labels are absent for any given sample. Standard binary cross-entropy treats positive and negative examples symmetrically, causing the loss to be dominated by the large number of easy negative samples and suppressing gradient signal from the rare positive examples. ASL addresses this through two mechanisms: an asymmetric focusing parameter that applies different hard-example mining strengths to positive and negative samples, and a probability margin that shifts the decision boundary by discarding very

confident negative predictions entirely. Specifically, the negative focusing parameter γ^- down-weights well-classified negatives aggressively, whilst the positive focusing parameter γ^+ is set to zero to preserve all positive gradient signal. In this project, ASL is used as the primary training loss with $\gamma^- = 4.0$ and $\gamma^+ = 0.0$, addressing the multi-label class imbalance inherent in the 3DSSG relation schema — where directional relations such as higher_than occur hundreds of thousands of times after injection, whilst contact relations such as on_top_of remain comparatively rare. ASL is one of the key contributors to the performance jump from Macro F1 = 0.52 (standard BCE) to Macro F1 = 0.919 in the final GeoKAN-RBF model.

Wu et al. (2020) — Tackling the Unannotated

Wu et al. examined the problem of annotation bias in scene graph generation datasets, demonstrating that standard supervised training on sparsely annotated scene graph datasets systematically biases models towards the small subset of labelled relationships whilst ignoring the majority of valid but unannotated pairs. The authors showed that treating unannotated pairs as true negatives — the default assumption in standard cross-entropy training — introduces a strong false negative signal that suppresses recall on valid but unlabelled relationships. They proposed bias-reduced training strategies including re-weighting and semi-supervised label propagation to mitigate this effect. Whilst this project does not directly implement the specific methods proposed by Wu et al., their analysis directly motivates the rule-based label injection strategy employed here: rather than treating the approximately 90 per cent of unannotated directed pairs in 3DSSG as true negatives, geometric rules are used to derive confident pseudo-labels for pairs whose spatial relationships are deterministically derivable from centroid positions. This converts the false negative problem into a supervised signal, and is the primary driver of the improvement from Macro F1 = 0.505 (GeoKAN-v1, without injection) to Macro F1 = 0.926 (GeoKAN-Gamma, with injection).

2.8 Clustering

McInnes et al. (2017) — HDBSCAN

McInnes et al. introduced HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), a density-based clustering algorithm that extends DBSCAN by converting it into a hierarchical framework and extracting a flat partition from the resulting cluster hierarchy using an excess of mass criterion. Unlike DBSCAN, which requires a fixed neighbourhood radius ε as input, HDBSCAN operates across all density scales simultaneously, building a minimum spanning tree over the data and extracting stable clusters based on their persistence across the density hierarchy. This makes HDBSCAN substantially more robust to varying local density — a critical property for clustering Gaussian splats, where object density varies considerably between compact objects such as a watch and large objects such as a router or box. A further advantage is HDBSCAN's explicit handling of noise: points that do not belong to any sufficiently dense cluster are assigned a noise label and excluded from object instances, preventing low-opacity floaters or background remnants from forming spurious clusters. In this project,

HDBSCAN is applied to a 5-dimensional feature matrix of XYZ position and RGB colour for each Gaussian, normalised using StandardScaler, with min_cluster_size auto-tuned by sweeping a set of divisors and selecting the value that produces a cluster count within the target range for each scene. An Otsu saturation threshold is applied prior to clustering to suppress achromatic background Gaussians that survive opacity filtering. This fully parameter-free, scene-adaptive clustering pipeline is applied identically to all scenes — training, LERF evaluation, and tabletop cross-domain evaluation — with no scene-specific tuning.

2.9 Research Gap and Positioning of this Work

The literature reviewed in the preceding sections reveals a consistent pattern in the evolution of 3D scene graph generation: as the quality of 3D scene representations has improved — from RGB-D depth maps to neural radiance fields to 3D Gaussian Splats — the methods used to infer spatial relations have grown increasingly dependent on large foundation models. GaussianGraph (Li et al., 2025) combines CLIP, SAM, LLaVA-1.6, and GroundingDINO. ReLaGS (Arafa et al., 2026) distils language features from vision-language models into Gaussian primitives before applying graph-based relational reasoning. ConceptGraphs (Gu et al., 2024) uses an LLM to infer relations from natural language object descriptions. Open3DSG (Koch et al., 2024) co-embeds 3D features with CLIP to achieve open-vocabulary prediction. The implicit assumption shared across all of these methods is that understanding spatial relations in a 3D scene requires semantic grounding — that a model must know what objects are before it can reason about how they relate.

This assumption has not been systematically challenged. None of the reviewed methods investigate whether the geometric signal available in a Gaussian splat — the positions, scales, orientations, and covariance structures of individual Gaussian primitives — is itself sufficient for accurate spatial relation prediction, independently of any semantic label. The question is non-trivial: object identity is not required to determine that one object rests on top of another, that two objects are adjacent, or that one is to the left of another. These relationships are geometric facts derivable from cluster centroids and bounding volumes alone.

A second gap concerns annotation efficiency. The 3DSSG dataset, which is the primary benchmark for 3D indoor scene graphs, leaves approximately 90 per cent of valid directed object pairs unannotated. No prior work on 3DSSG has directly addressed this sparsity through deterministic geometric label injection — the systematic derivation of pseudo-labels for unannotated pairs using rules that are provably correct for directional relations.

A third gap concerns post-prediction consistency. Scene graph outputs from all reviewed methods can contain logical contradictions — for example, predicting both higher_than(A, B) and higher_than(B, A) simultaneously, or on_top_of(A, B) without the corresponding under(B, A). None of the reviewed methods apply a formal, zero-

parameter constraint enforcement mechanism to resolve such contradictions after prediction.

This project addresses all three gaps. It proposes a geometry-first pipeline that makes no use of foundation models at inference time, resolves annotation sparsity through rule-based label injection, and enforces logical consistency through a zero-parameter symbolic repair module. The result is a system that is simultaneously more accurate, more efficient, and more logically coherent than existing Gaussian-based scene graph methods.

2.10 Summary Table

Reference	Year	Venue	Key Contribution	Role in LogicSplat
Kerbl et al.	2023	SIGGRAPH	3D Gaussian Splatting — explicit radiance field with per-Gaussian covariance	Core scene representation; source of all geometric features
Tancik et al.	2023	SIGGRAPH	NerfStudio — modular NeRF/3DGS training framework	Reconstruction engine (splatfacto) used for all scene inputs
Kerr et al.	2023	ICCV	LERF — language-embedded radiance fields	Source of 4 held-out evaluation scenes
Wald et al.	2019	ICCV	3RScan — large-scale indoor re-scan dataset	Primary training and validation dataset (480+85 scenes)
Wald et al.	2020	CVPR	3DSSG — 3D semantic scene graph dataset and benchmark	Primary benchmark; annotation source for training labels
Dai et al.	2017	CVPR	ScanNet — richly annotated indoor RGB-D dataset	Used in exploratory baseline phase; not in final training
Wu et al.	2021	CVPR	SceneGraphFusion — incremental online scene graph prediction	Establishes GNN-based 3D relational reasoning baseline
Koch et al.	2024	CVPR	Open3DSG — open-vocabulary 3D scene graphs from point clouds	Comparison baseline; R@3=0.580 vs ours 0.878

Reference	Year	Venue	Key Contribution	Role in LogicSplat
Gu et al.	2024	ICRA	ConceptGraphs — LLM-based open-vocabulary 3D scene graphs	Comparison baseline; R@3=0.740 vs ours 0.878
Qin et al.	2024	CVPR	LangSplat — language-feature-embedded Gaussian splats	Contextualises semantic Gaussian understanding; motivates geometry-only alternative
Ye et al.	2024	ECCV	Gaussian Grouping — SAM-supervised Gaussian instance segmentation	Contrasts foundation-model segmentation with geometry-only HDBSCAN clustering
Li et al.	2025	arXiv	GaussianGraph — scene graph generation from 3DGS with foundation models	Primary LERF baseline; mR@5=63.2% vs ours 92.9%
Arafa et al.	2026	CVPR	ReLaGS — language-distilled Gaussian scene graph without scene-specific training	State-of-the-art comparison; R@3=0.790, 1.5B params vs ours 1.018M
Veličković et al.	2018	ICLR	GAT — graph attention networks with learnable neighbourhood weighting	Foundational GNN architecture; used in baseline models v1–v7
Brody et al.	2021	ICLR	GATv2 — dynamic graph attention with full expressive power	Message passing backbone of GeoKANRelationGNN
Liu et al.	2024	ICLR	KAN — Kolmogorov-Arnold Networks with learnable edge activations	Theoretical foundation of the GeoKAN layer
Sen et al.	2026	arXiv	GeoKAN — geometry-adaptive KAN with diagonal Riemannian metric	Direct architectural basis for GeoKAN-Gamma and GeoKAN-RBF
Ridnik et al.	2021	ICCV	Asymmetric Loss — asymmetric focal loss for multi-label imbalance	Training loss; key contributor to F1 improvement from 0.52 to 0.919

Reference	Year	Venue	Key Contribution	Role in LogicSplat
Wu et al.	2020	arXiv	Tackling the Unannotated — annotation bias in scene graph datasets	Motivates rule-based label injection strategy
McInnes et al.	2017	JOSS	HDBSCAN — hierarchical density-based clustering with noise handling	Object clustering algorithm with Otsu auto-tuning across all scenes

Table 2.1: Summary of reviewed literature and their role in the LogicSplat project.

3.Methodology

3.1 System Overview and Pipeline Architecture

LogicSplat is a geometry-first pipeline for spatial scene graph generation from 3D Gaussian Splats. The system takes as input a short smartphone video of an indoor scene and produces as output a directed scene graph whose nodes represent segmented object instances and whose edges represent predicted spatial relations from a fixed vocabulary of ten relation types: on_top_of, under, attached_to, adjacent_to, left_of, right_of, in_front_of, behind, higher_than, and lower_than. No foundation model, vision-language model, or depth sensor is required at inference time.

The pipeline consists of eight sequential stages, illustrated in Figure 3.1:

Stage 1 — Scene Reconstruction. A smartphone video of approximately 60 frames at 640×480 resolution is processed by COLMAP to estimate camera poses via structure-from-motion. The recovered poses and images are passed to NerfStudio splatfacto, which trains a 3D Gaussian Splat over 30,000 iterations and exports the result as a .ply file containing all Gaussian primitives with their positions, colours, opacities, scales, and rotations.

Stage 2 — Gaussian Preprocessing. The raw Gaussian cloud is cleaned in two steps: an opacity filter removes Gaussians below a threshold of 0.1, and Statistical Outlier Removal (SOR) discards spatially isolated Gaussians whose mean distance to their 20 nearest neighbours exceeds two standard deviations above the scene mean. A RANSAC-based plane segmentation pass then identifies and removes the dominant horizontal surface — the floor or table — to isolate the object region.

Stage 3 — Object Clustering. The cleaned Gaussian cloud is passed to an HDBSCAN clustering pipeline operating on a 5-dimensional feature matrix of XYZ position and normalised RGB colour. The min_cluster_size parameter is auto-tuned per scene using an Otsu saturation threshold and a divisor sweep targeting the expected number of objects. Each resulting cluster becomes an Object3D instance with a centroid, bounding box, and point count.

Stage 4 — Feature Extraction. For each object instance, a 10-dimensional node feature vector is computed from the cluster's centroid, bounding box dimensions, volume, height ratio, point density, and ordinal height rank — all normalised to be scale-invariant. For each directed pair of objects, a 22-dimensional edge feature vector is computed encoding directional offsets, 3D distance, bounding box overlap, volume ratio, vertical gap, and composite contact scores.

Stage 5 — Relation Prediction. The object graph, with node and edge features as above, is passed to GeoKANRelationGNN. Two GATv2Conv message passing layers with four attention heads propagate geometric context across the graph. A pair projection head concatenates source and target node embeddings into a 512-dimensional pair representation. Two GeoKAN classification heads — one for contact relations and one for directional relations — produce ten scalar logits per directed object pair.

Stage 6 — Threshold Calibration. Each of the ten relation logits is thresholded independently using per-relation thresholds calibrated on the 3DSSG validation set. This converts continuous logits into binary relation predictions.

Stage 7 — Symbolic Consistency Repair. The predicted relation set is passed to SceneGraphRepair, a zero-parameter module that enforces four categories of logical constraints — inverse completeness, mutual exclusion, asymmetry, and transitivity — through fixed-point iteration until the graph reaches a contradiction-free state.

Stage 8 — Output. The final scene graph is returned as a set of directed relation triples of the form (subject, relation, object), suitable for downstream tasks including spatial querying, robotic planning, and scene summarisation.

The complete pipeline from a cached Gaussian splat to a final scene graph executes in under two seconds on an NVIDIA RTX 3060 GPU using less than 1.5 GB of GPU memory.

3.2 Scene Reconstruction Pipeline

3.2.1 Smartphone Video Capture

Scene reconstruction begins with a monocular RGB video captured using a consumer smartphone camera. For the eight custom tabletop evaluation scenes collected for this project, videos were recorded at 640×480 resolution at approximately 30 frames per second, with the camera moved in a slow arc around the scene to ensure adequate multi-view coverage of all objects. Approximately 60 frames were extracted per scene, selected to provide uniform angular coverage without excessive redundancy between adjacent frames. No depth sensor, structured light projector, or calibration target was used. This capture protocol was chosen deliberately to reflect a realistic deployment scenario: the

entire pipeline, from video capture through scene graph generation, should be executable by an end user with no specialist equipment.

3.2.2 COLMAP Structure-from-Motion

Camera pose estimation is performed using COLMAP, a general-purpose structure-from-motion and multi-view stereo pipeline. COLMAP detects SIFT keypoints in each frame, matches them across frame pairs using approximate nearest-neighbour search, and recovers per-frame camera poses and a sparse 3D point cloud through incremental bundle adjustment. The recovered camera intrinsics and extrinsic poses are stored in a transforms.json file that NerfStudio uses to initialise the Gaussian training process. For the 3RScan training scenes, pre-computed COLMAP poses from the GaussianWorld dataset are used directly, as the original 3RScan sequences were captured with a Structure Sensor whose poses are already available. For the custom tabletop and LERF evaluation scenes, COLMAP is run from scratch on the captured frames.

3.2.3 NerfStudio Splatfacto

Given camera poses and images, NerfStudio's splatfacto pipeline trains a 3D Gaussian Splat using the method of Kerbl et al. (2023). Training proceeds for 30,000 iterations using the Adam optimiser, with periodic densification steps that split or clone Gaussians in high-gradient regions and pruning steps that remove Gaussians with low opacity or excessive scale. The output is a .ply file containing all surviving Gaussian primitives, each parameterised by position (x, y, z), colour as spherical harmonic DC coefficients (f_dc_0, f_dc_1, f_dc_2), opacity stored as a logit value, log-scale per axis (scale_0, scale_1, scale_2), and rotation as a quaternion (rot_0, rot_1, rot_2, rot_3). Reconstruction time on an NVIDIA RTX 3060 is approximately 5–15 minutes per scene depending on scene complexity, representing a one-time cost per scene after which all subsequent pipeline stages operate on the fixed .ply file. For the 3RScan training scenes, pre-trained Gaussian splats from the GaussianWorld MCMC repository are used directly, avoiding the need to retrain splats for all 565 training and validation scenes.

3.3 Datasets

3.3.1 3RScan and 3DSSG Annotations

The primary training and validation data for this project is drawn from two complementary sources: the 3RScan scan dataset (Wald et al., 2019) and the 3D Semantic Scene Graph (3DSSG) annotation layer (Wald et al., 2020). 3RScan provides 1,482 indoor RGB-D scan sequences of 478 distinct environments, each reconstructed as a textured mesh with per-vertex instance segmentation labels. 3DSSG provides human-annotated pairwise spatial relationships between object instances across these scenes, using a relation vocabulary of 26 types. For this project, the original 26-type vocabulary is reduced to 10 relations by aggregating semantically equivalent predicates — for example, standing on, supported by, and lying on are all mapped to on_top_of — and dropping two relations with insufficient positive examples: INSIDE (262 positives, 0.038 per cent positive rate) and HANGING_FROM (974 positives, 0.14 per cent positive rate).

Gaussian splat reconstructions of 565 3RScan scenes are sourced from the GaussianWorld repository, which provides MCMC-based 3DGS reconstructions of the original sequences. Of these, 565 scenes are used in total: 480 for training and 85 for validation, with a scene-level random split at seed 42 ensuring no scene appears in both splits. The 46 RIO10 scenes are excluded from training as they form a held-out benchmark for re-localisation tasks unrelated to this project. Scenes with fewer than three object instances or less than 50 per cent Gaussian coverage after instance label transfer are excluded, yielding final training and validation sets of 480 and 85 scenes respectively. The training set contains a total of 1,005,126 directed object pair edges, expanding to 2,383,223 positive label instances after rule-based injection.

3.3.2 Instance Label Transfer to Gaussian Splats

The 3DSSG relation annotations reference object instances defined on 3RScan mesh vertices. To transfer these instance identities to the Gaussian splat representation, each Gaussian primitive is assigned the instance label of its nearest labelled mesh vertex using a nearest-neighbour search in 3D Euclidean space. Gaussians that do not correspond to any labelled instance are assigned a background label of −1 and excluded from the object graph. This transfer achieves a mean coverage of 82.7 per cent across the 565 scenes, meaning that on average 82.7 per cent of Gaussians in each scene receive a valid non-background instance label. The 3DSSG object identifiers correspond directly to 3RScan instance IDs, so no additional object matching step is required.

3.3.3 ScanNet (Exploratory Phase)

ScanNet (Dai et al., 2017) was used during the early exploratory phase of model development, specifically for training GAT-based baseline architectures (model versions v1 through v7) prior to the transition to 3RScan. During this phase, spatial relation labels were derived entirely from geometric rules applied to cluster centroids, as ScanNet does not provide scene graph annotations in the 3DSSG format. Whilst the ScanNet-trained models achieved acceptable in-domain performance, they failed to generalise to the custom tabletop evaluation scenes due to the significant scale and appearance gap between ScanNet room-scale indoor reconstructions and the tabletop environment. This cross-domain failure motivated the decision to train the final GeoKAN models on 3RScan with human-annotated 3DSSG labels rather than geometry-derived pseudo-labels alone. ScanNet data is not used in any aspect of the final GeoKAN model training or evaluation.

3.3.4 Custom Tabletop Scenes (Cross-Domain Evaluation)

Eight tabletop scenes (scene_06 through scene_13) were captured and reconstructed specifically for cross-domain evaluation. Each scene contains four or five everyday objects — including a Wi-Fi router, a pen, a water bottle, an Agaro hair dryer box, a watch, a perfume bottle, and a cream tub — arranged on a flat surface at varying positions and with varying stacking configurations. Five scenes include a router physically placed on top of the Agaro box; three scenes have all objects placed side by side without stacking. Scene extents range from 0.3 to 0.8 metres — approximately one order

of magnitude smaller than the 3–8 metre room-scale scenes used for training. Three scenes (scene_09, scene_12, scene_13) were reconstructed with an inverted Z-axis convention, where more negative Z values correspond to physically higher positions; this is handled during ground truth construction by negating Z coordinates before deriving spatial relations. Ground truth relation files for all eight scenes were constructed programmatically using build_gt_all_scenes.py, which runs the same deterministic HDBSCAN clustering pipeline used during evaluation, reads the actual cluster centroids, and derives all pairwise spatial relations geometrically from those centroids. Cluster-to-object assignments were verified by visual inspection of centroid positions and point counts.

3.3.5 LERF Scenes (Held-Out Evaluation)

Four scenes from the LERF dataset (Kerr et al., 2023) — ramen, teatime, waldo_kitchen, and figurines — are used as a held-out evaluation set to assess in-domain generalisation to an unseen dataset. These scenes cover indoor environments at 1–5 metre scale, which is comparable to the 3–8 metre training distribution of 3RScan, making them a test of generalisation to a different dataset rather than a different scale domain. Each scene contains 10–13 object instances and was originally captured with over 100 frames per scene for the LERF paper. Ground truth relation triples for all four scenes are derived using the same deterministic geometric derivation approach as for the tabletop scenes, yielding 1,770 ground truth triples in total across the four scenes. A random sample of these triples was manually verified for correctness prior to evaluation.

3.4 Gaussian Splatting Data Pipeline

3.4.1 Loading 3DGS .ply Files

The Gaussian splat is loaded from a standard 3DGS-format .ply file using the plyfile library. Each vertex in the .ply file corresponds to one Gaussian primitive and carries the following attributes: position (x, y, z), spherical harmonic DC colour coefficients (f_dc_0, f_dc_1, f_dc_2), opacity stored as a raw logit, log-scale per axis (scale_0, scale_1, scale_2), and rotation as a quaternion (rot_0, rot_1, rot_2, rot_3). RGB colour values are recovered from the DC spherical harmonic coefficients using the zero-order SH basis constant $C_0 = 0.28209479177387814$, giving RGB = $(f\_dc \cdot C_0 + 0.5)$ clipped to [0, 1]. Opacity values are converted from logit to probability via the sigmoid function: opacity = $1 / (1 + \exp(-\text{logit}))$. The full 3×3 covariance matrix for each Gaussian is computed from its scale and rotation by constructing the rotation matrix from the normalised quaternion, forming the scale matrix $S = \text{diag}(\exp(\text{log\_scale}))$, and computing $\Sigma = RS \cdot (RS)^T$. The upper triangle of Σ — six values — is stored per Gaussian. Any Gaussian primitive containing NaN or infinite values in position, opacity, or scale is removed with a warning before further processing.

3.4.2 Opacity Filtering and Statistical Outlier Removal

Two sequential filtering operations are applied to the loaded Gaussian cloud before clustering. First, an opacity filter removes all Gaussians with opacity below a threshold of 0.1, eliminating transparent background Gaussians and low-confidence primitives that were retained by the splat training process but carry no meaningful geometric information. Second, Statistical Outlier Removal (SOR) is applied using Open3D's remove_statistical_outlier function with nb_neighbors = 20 and std_ratio = 2.0. SOR computes for each Gaussian the mean Euclidean distance to its 20 nearest neighbours; any Gaussian whose mean distance exceeds the scene-wide mean plus 2.0 standard deviations is classified as an isolated floater and removed. This step eliminates spatially isolated Gaussian primitives that survive opacity filtering — typically artefacts of the splat training process that cluster away from the main scene geometry. A scikit-learn nearest-neighbours fallback is available in environments where Open3D is not installed. Together, these two filters reduce the Gaussian count to the geometrically coherent core of the scene before background removal.

3.4.3 Background Removal via RANSAC Plane Segmentation

Following opacity and outlier filtering, a RANSAC-based plane segmentation step removes the dominant horizontal surface — the floor in room-scale scenes or the table surface in tabletop scenes — to isolate the objects of interest. The algorithm samples up to 50,000 Gaussians from the filtered cloud, constructs an Open3D PointCloud object, and runs segment_plane with a distance threshold of 1 per cent of the scene bounding box diagonal, three RANSAC seed points, 1,000 iterations, and a fixed random seed of 42 for reproducibility. The plane equation (a, b, c, d) returned by RANSAC defines the dominant surface; the signed distance of each Gaussian from this plane is computed as $h = (xyz \cdot \hat{n}) + d/|n|$, where $\hat{n}$ is the unit plane normal. To determine which side of the plane contains the objects rather than the floor or ceiling, the 95th percentile of positive and negative signed distances is compared; the side with the smaller spread is identified as the object side. Gaussians within the object region — specifically those with height in $[-\text{margin}, h_{97}]$ where margin = 5 per cent of h_{97} — are retained, and all others are removed. If fewer than 10 per cent of Gaussians pass this filter, the step is skipped and the unmodified cloud is passed downstream to prevent catastrophic data loss on unusual scene geometries.

3.5 Object Clustering

3.5.1 HDBSCAN with Auto-Tuning

Following background removal, the remaining Gaussians are clustered into discrete object instances using HDBSCAN (McInnes et al., 2017). The clustering operates on a 5-dimensional feature matrix constructed by concatenating XYZ position (3 dimensions) and RGB colour normalised to [0, 1] and weighted by a factor of 0.3 (3 dimensions), yielding a combined spatial-colour representation that groups Gaussians both by proximity and by appearance. The feature matrix is standardised using StandardScaler before clustering, ensuring that the spatial and colour dimensions contribute comparably

to the distance metric. HDBSCAN is run with min_samples = 3 and cluster_selection_method set to either eom or leaf, with both tried during auto-tuning.

The primary challenge with HDBSCAN for this application is that min_cluster_size — the minimum number of points for a group to be considered a cluster — must be set appropriately for each scene, as Gaussian splat density varies considerably across scenes and objects. A fixed value produces either over-segmentation in sparse scenes or under-segmentation in dense ones. To address this, min_cluster_size is auto-tuned per scene by sweeping a set of divisors [3, 5, 8, 10, 15, 20, 30, 50, 80, 100, 150, 200, 300, 500] of the total Gaussian count subsampled to 6,000 points, running HDBSCAN at each divisor, and selecting the value that produces a cluster count closest to the midpoint of the target range [target_min, target_max]. For training scenes, the target range defaults to [4, 12]; for evaluation scenes, the expected object count is passed as n_exact, which sets both target_min and target_max to that value, forcing the auto-tuner to find the min_cluster_size that produces exactly the expected number of clusters. Gaussians not included in the 6,000-point subsample are assigned to their nearest cluster centroid in feature space via L2 distance after the initial clustering is complete.

3.5.2 Otsu Saturation Thresholding

Prior to HDBSCAN clustering, an optional Otsu saturation threshold is applied to suppress achromatic background Gaussians — walls, floors, and ceiling remnants — that survive the RANSAC plane removal step. The saturation of each Gaussian is computed as sat = max(R, G, B) − min(R, G, B) from its normalised RGB values. A 256-bin histogram of saturation values is computed and Otsu's method is applied to find the threshold that maximises inter-class variance between low-saturation (background) and high-saturation (object) Gaussians. Gaussians below the Otsu threshold are removed before clustering. If fewer than 5 per cent of Gaussians pass the threshold — indicating a scene where all objects are achromatic — the threshold is halved and re-applied; if still fewer than 5 per cent pass, the filter is disabled entirely and all Gaussians are passed to clustering. This fallback prevents the saturation filter from destroying valid data in scenes containing predominantly grey or white objects.

3.5.3 Z-Axis Correction After Clustering

A coordinate system correction is applied after clustering to ensure consistent spatial reasoning across all scenes. The 3DGS training convention used by NerfStudio splatfacto defines the camera coordinate system with −Z pointing upward, meaning that physically higher objects have more negative Z coordinates in the raw splat. After clustering, the Z coordinate of all Object3D centroids and bounding box extremes is negated, converting the coordinate system to +Z = up. This correction is applied uniformly to all scenes processed through the standard pipeline. The three tabletop evaluation scenes that exhibit Z-DOWN orientation (scene_09, scene_12, scene_13) were handled separately during ground truth construction, where the SCENE_Z_FLIP flag in build_gt_all_scenes.py ensures that effective height is computed as −Z for those scenes. All subsequent feature computation and relation derivation assumes the +Z = up convention.

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3.6 Feature Engineering

3.6.1 Node Features (10-Dimensional)

Each object instance is represented by a 10-dimensional node feature vector computed entirely from the geometric properties of its Gaussian cluster. All features are designed to be scale-invariant, normalised relative to scene-level statistics rather than absolute metric values, so that a model trained on room-scale 3RScan scenes can be applied to tabletop scenes without encountering out-of-distribution absolute magnitudes. The normalisation divisor used throughout is the average bounding box diagonal across all object instances in the scene, denoted avg_diag.

Features 0–2 encode the normalised centroid position (c_x − scene_min) / scene_mean_diag, clipped to a valid range, providing the object's location within the scene coordinate frame. Features 3–5 encode size aspect ratios computed as size_axis / self_diagonal, where self_diagonal is the object's own bounding box diagonal; these capture the object's shape (flat, elongated, or compact) independently of its absolute size. Feature 6 is a log-normalised volume: log(volume / scene_median_volume + 1) / 5, clipped to [0, 1], providing a relative measure of object size within the scene. Feature 7 is the height ratio size_z / max(size_x, size_y), encoding vertical elongation — distinguishing tall objects such as bottles from flat objects such as books. Feature 8 is point density, computed as point_count / volume and normalised to [0, 1] across the scene, reflecting how densely the object is represented in the Gaussian splat. Feature 9 is the ordinal Z-rank of the object's centroid among all scene objects, normalised to [0, 1]; this completely scale-invariant feature encodes relative vertical position and is one of the strongest signals for higher_than and lower_than prediction.

3.6.2 Edge Features (22-Dimensional)

For each directed object pair (A → B), a 22-dimensional edge feature vector encodes the geometric relationship between the two objects. The normalisation divisor for edge features is avg_diag = (diag_A + diag_B) / 2, the mean of both objects' bounding box diagonals.

Features 0–2 encode directional offsets: delta_x = (c_x_A − c_x_B) / avg_diag, delta_y = −(c_y_A − c_y_B) / avg_diag (negated to match the 3DSSG convention where −Y is the front direction), and delta_z = (c_z_A − c_z_B) / avg_diag, all clipped to [−3, 3] and divided by 3. These three features are the primary signals for left_of, right_of, in_front_of, behind, higher_than, and lower_than. Feature 3 is the XY planar distance and Feature 4 is the full 3D Euclidean distance between centroids, both normalised by avg_diag. Feature 5 is the XY bounding box footprint overlap of A divided by A's footprint area. Feature 6 is the volume ratio min(vol_A, vol_B) / max(vol_A, vol_B). Feature 7 is the log height ratio log(size_z_A / size_z_B) clipped to [−3, 3] / 3.

Features 8–16 are composite contact and support signals specifically designed for on_top_of and under prediction. Feature 8 (vert_gap) encodes the signed vertical gap between A's bottom face and B's top face, normalised by the average object height; a small positive value indicates A is resting directly on B. Features 9–11 provide alternative normalisations of the same vertical gap. Feature 12 (contact_score) is a composite signal: exp(−|gap| / threshold) × bbox_overlap × (A_above_B), combining proximity, footprint overlap, and vertical ordering into a single contact confidence value. Features 13–16 encode XY footprint overlap relative to B's area, proximity to the on_top_of rule boundary, distance from B's footprint edge, and a stacking-versus-side-by-side discriminator (Δz − 0.5·Δxy) / max(|Δz|, ε).

Features 17–21 provide containment signals. Feature 17 is the 3D volumetric overlap divided by A's volume. Feature 18 is a binary flag indicating whether A's centroid lies inside B's bounding box. Features 19–21 encode the vz_ratio Δz / max(xy_dist, 0.05·avg_diag) and XY containment fractions along each axis.

3.6.3 Scale-Invariant Normalisation Strategy

All node and edge features are normalised relative to scene-level statistics — specifically the average object bounding box diagonal — rather than in absolute metric units. This design choice is the primary mechanism enabling cross-domain transfer: a model trained on 3RScan rooms where avg_diag ≈ 0.8–1.2 m will encounter similar normalised feature values on tabletop scenes where avg_diag ≈ 0.15–0.25 m, because both are expressed as multiples of their respective scene's average object size. Without this normalisation, the absolute centroid offsets and distances in the tabletop scenes would fall far outside the training distribution, causing catastrophic prediction failure. The ordinal Z-rank feature (node feature 9) takes this principle to its logical extreme, encoding only the rank order of objects by height and discarding all metric information entirely.

3.7 Rule-Based Label Injection

3.7.1 Motivation: Annotation Sparsity in 3DSSG

The 3DSSG dataset, whilst the most comprehensive source of human-annotated 3D spatial relations available, suffers from severe annotation sparsity. Human annotators labelled between 80 and 120 directed object pairs per scene, out of approximately 870 possible directed pairs in a scene with around 30 objects (N × (N−1) directed pairs). This

means that approximately 90 per cent of geometrically valid object pair relationships are left unannotated. Standard supervised training treats unannotated pairs as true negatives — that is, as evidence that no relation holds — which introduces a pervasive false negative signal. For directional relations such as left_of, this is particularly damaging: if object A is to the left of object B, this is a geometric fact that holds regardless of whether a human annotator chose to label it. Training a model to predict left_of = 0 for this pair because it was not annotated actively teaches the model an incorrect spatial fact. The consequence is visible in the pre-injection baseline: GeoKAN-v1, trained on 3DSSG labels alone, achieves only 40,000 positive examples each for left_of and right_of — a tiny fraction of the true positive count — resulting in Macro F1 = 0.505 despite strong ranking performance.

3.7.2 Directional Relation Rules (Confidence = 1.0)

Six of the ten relations — left_of, right_of, in_front_of, behind, higher_than, and lower_than — are fully determined by the relative positions of object centroids and carry no ambiguity. For any directed pair (A, B), the following rules are applied with confidence 1.0 to all unannotated pairs:

- left_of(A, B) if c_x_A < c_x_B
- right_of(A, B) if c_x_A > c_x_B
- in_front_of(A, B) if c_y_A < c_y_B (3DSSG convention: −Y is front)
- behind(A, B) if c_y_A > c_y_B
- higher_than(A, B) if c_z_A > c_z_B
- lower_than(A, B) if c_z_A < c_z_B

After injection, the directional relation counts increase from approximately 40,000 per class to between 207,603 (in_front_of, behind) and 450,434 (left_of, right_of) positive examples, representing an 8–11 times increase in training signal. These injected labels are geometric facts — they cannot be wrong given correct centroid positions — and are therefore assigned confidence 1.0 with no down-weighting.

3.7.3 Contact and Support Rules

The four contact relations — on_top_of, under, attached_to, and adjacent_to — are not fully determined by centroid positions alone and are injected with lower confidence to account for annotation inconsistency and rule imprecision.

on_top_of(A, B) and under(B, A) are injected when A's bottom bounding face is within a scene-adaptive vertical threshold of B's top bounding face, A's XY footprint overlaps B's footprint, and A's centroid is above B's centroid. These conditions are evaluated using the vert_gap and contact_score edge features, with injection confidence 0.75.

adjacent_to(A, B) is injected when the 3D Euclidean distance between centroids is below 0.8 times the average object diagonal. Due to the inherent ambiguity of the adjacent_to label — annotators apply it inconsistently for near-equal-distance pairs — injection confidence is set to 0.55, the lowest of any injected relation.

attached_to(A, B) is injected when the bounding box gap between A and B is less than 5 per cent of the average object diagonal, indicating physical contact consistent with wall-mounted or built-in object configurations. Injection confidence is 0.60.

3.7.4 Inverse Label Injection (Fix2)

A separate injection step addresses a systematic gap in the 3DSSG annotations: the under relation is almost never explicitly annotated, because annotators tend to label the relationship from the perspective of the supported object (on_top_of) rather than the supporting one. This results in near-zero positive examples for under in the raw 3DSSG labels, causing any model trained without correction to achieve essentially zero recall for under.

Fix2 resolves this by injecting the logical inverse of every explicitly annotated relation. For every 3DSSG-annotated triple (A, R, B), the inverse triple (B, R_inv, A) is injected with the same confidence, where the inverse relation pairs are: on_top_of ↔ under, higher_than ↔ lower_than, left_of ↔ right_of, in_front_of ↔ behind. Human labels always take priority — Fix2 never overwrites an existing annotation — but it fills the systematic gaps left by annotators who labelled only one direction of a symmetric or inverse relationship.

3.7.5 Confidence Weighting and Priority Scheme

All injected labels are represented as soft labels rather than hard binary targets. A confidence value $c \in \{0.55, 0.60, 0.75, 1.0\}$ is assigned per relation type as described above; the target for that relation in the loss function is set to c rather than 1.0, effectively applying label smoothing proportional to the uncertainty of the rule. The priority scheme ensures that human-annotated 3DSSG labels always override injected labels: if a directed pair (A, B) has an explicit 3DSSG annotation for relation R, the injected label for R on that pair is discarded and the human annotation is used with confidence 1.0. This preserves the full value of human annotation whilst extending coverage to the 90 per cent of unannotated pairs.

3.8 Model Architecture: GeoKANRelationGNN

3.8.1 The GeoKAN Layer

The GeoKAN layer is the core computational unit of the classification heads in GeoKANRelationGNN. It implements a geometry-adaptive Kolmogorov-Arnold Network in which basis expansion is performed in learned, warped coordinates rather than the raw input space. A single GeoKAN layer proceeds through five steps.

Step 1 applies Batch Normalisation to the input vector u, producing a normalised representation u_normed. Step 2 computes a diagonal geometric metric $g \in \mathbb{R}^d$, where d is the input dimension. In the GeoKAN-Gamma variant, g is computed as g_i =

softplus(γ_i) + 1e-6, where γ is a learnable parameter vector initialised to zero, giving an initial metric of approximately 1.0 (identity scaling) at the start of training. In the GeoKAN-RBF variant, a small MetricNet — a two-layer MLP of structure Linear(d→64)→GELU→Linear(64→d) — computes g as softplus(MetricNet(u_normed)) + 1e-6, clamped to [0.05, 16.0], producing an input-dependent adaptive metric. Step 3 computes the warped representation z = u_normed × √g, stretching the normalised input by the square root of the learned metric. Step 4 applies a radial basis function expansion: for each of K = 12 learnable centres c_k spaced linearly in [−3, 3], the activation is φ_k = exp(−γ_rbf × (z − c_k)^2), where γ_rbf = softplus(learnable_scalar). In the GeoKAN-Wavelet variant, the Mexican hat wavelet φ_k = (1 − t^2) × exp(−t^2/2) with t = (z − c_k) is used in place of the RBF. Step 5 concatenates the flattened RBF activations with u_normed and passes the result through a linear projection followed by GELU activation and dropout, producing the output representation.

A metric regularisation term L_metric = 1e-4 × mean(log(g)^2) is added to the training loss to penalise extreme metric values and encourage the metric to remain near identity unless deviation is justified by the training signal.

3.8.2 Node Encoder

The node encoder transforms the 10-dimensional input node feature vector into a 128-dimensional embedding using a single linear layer followed by LayerNorm and GELU activation: h = GELU(LayerNorm(Linear(10 → 128))). This lightweight encoder contains 1,408 parameters and serves to project the heterogeneous node features into a uniform embedding space before graph attention message passing.

3.8.3 GATv2Conv Message Passing Layers

Two successive GATv2Conv layers propagate geometric context across the object graph. Each layer uses four attention heads, an input and output dimension of 128 (with heads averaged rather than concatenated, so the output dimension is also 128), and a 22-dimensional edge feature input that is incorporated into the attention coefficient computation. A residual connection adds the input embedding to the layer output before LayerNorm is applied. The two GATv2Conv layers together contain 148,736 parameters and allow each object node to aggregate relational context from its graph neighbourhood, incorporating information about the geometric relationships between connected objects via the edge features. After the second GATv2Conv layer, each node carries a 128-dimensional embedding that encodes both its own geometric properties and the context of its spatial relationships with other objects in the scene.

3.8.4 Pair Projection Head

For each directed object pair (A → B), a pair representation is constructed by concatenating four terms: the source node embedding h_A, the target node embedding h_B, their element-wise difference h_A − h_B, and their element-wise product h_A ⊙ h_B. This yields a 512-dimensional vector that captures both the individual properties of

each object and their relational contrast. A linear projection followed by LayerNorm and GELU reduces this to a 64-dimensional pair embedding: p = GELU(LayerNorm(Linear(512 → 64))). This head contains 33,024 parameters.

3.8.5 Contact GeoKAN Head

The contact head predicts four relations: on_top_of, under, attached_to, and adjacent_to. Its input is an 86-dimensional vector formed by concatenating the 64-dimensional pair embedding with the full 22-dimensional edge feature vector. Two sequential GeoKAN layers with hidden dimension 128 and $K = 12$ RBF centres process this input, followed by a linear output layer producing four logits. The contact head contains approximately 579,020 parameters in the RBF variant. The full edge feature vector is concatenated because contact relations depend critically on precise vertical gap and footprint overlap values that are directly present in the edge features.

3.8.6 Directional GeoKAN Head

The directional head predicts six relations: left_of, right_of, in_front_of, behind, higher_than, and lower_than. Its input is a 74-dimensional vector formed by concatenating the 64-dimensional pair embedding with the first 10 edge features — the directional offset, distance, and overlap features most relevant to directional prediction. Two GeoKAN layers with the same configuration as the contact head, followed by a linear output layer, produce six logits. The directional head contains approximately 301,531 parameters in the RBF variant. The separation into two heads allows each to specialise: the contact head attends to vertical proximity and footprint overlap, whilst the directional head focuses on centroid offsets and 3D distances.

3.8.7 Model Variants: Evolution from GAT Baseline to GeoKAN-Gamma

The final GeoKANRelationGNN architecture emerged through a systematic evolution spanning eleven model versions. The initial ScanNet-based models (v1–v3) used a single-head GAT with 8-dimensional node features and a softmax loss, achieving unstable Macro F1 ≈ 0.35. Successive improvements introduced multi-label binary cross-entropy (v4), corrected data leakage from augmentation before the train-validation split (v5), expanded edge features to include contact signals (v6), added the dual-head structure and asymmetric focal loss (v7). The transition to 3RScan with GeoKAN heads and real Gaussian data (GeoKAN-v1) achieved Macro F1 = 0.505 and R@5 = 0.967, but with a 12-class schema that included the near-zero INSIDE and HANGING_FROM relations. After scale-invariant feature normalisation (GeoKAN-v2), correction of a configuration error in the ASL parameters (GeoKAN-v3), and the introduction of rule-based label injection with the corrected 10-class schema (GeoKAN-RBF), Macro F1 jumped to 0.919. The final GeoKAN-Gamma variant replaced the adaptive MetricNet with a fixed diagonal metric, reduced the learning rate from 2×10^{-3} to 5×10^{-4}, and extended warmup from 5 to 10 epochs, achieving the best result of Macro F1 = 0.926 with 1,018,414 parameters — 53,504 fewer than GeoKAN-RBF.

3.9 Training Strategy

3.9.1 Asymmetric Loss

The primary training objective is the Asymmetric Loss (ASL) introduced by Ridnik et al. (2021), adapted for the multi-label spatial relation prediction task. Standard binary cross-entropy is ill-suited to this setting because the positive-to-negative ratio is highly skewed: even after rule-based label injection, the majority of directed object pairs carry no relation for most relation types, creating a large imbalance between positive and negative examples. ASL addresses this through asymmetric focusing: negative examples are down-weighted by a factor of $(1 - p)^{\gamma^-}$ that suppresses the gradient from well-classified negatives, whilst positive examples receive no focusing ($\gamma^+ = 0.0$) to preserve all positive gradient signal. In this project, $\gamma^- = 4.0$ and $\gamma^+ = 0.0$ are used, with no probability margin shift. This configuration was arrived at after correcting an earlier error in GeoKAN-v3, where $\gamma^+ = 4.0$ was mistakenly applied, causing training instability and degraded performance.

The total training loss has three components: L = L_ASL + L_inv + L_metric. The inverse consistency loss L_inv = 0.1 × MSE(p(R(A,B)), p(R_inv(B,A))) penalises disagreement between the predicted probability of a relation and the predicted probability of its inverse on the reversed pair, encouraging the model to learn logically consistent predictions during training. The metric regularisation loss L_metric = 1e-4 × $\mathrm{mean}(\log(g)^2)$ prevents the GeoKAN metric from collapsing to extreme values.

3.9.2 Per-Relation Loss Weights and Label Smoothing

Four contact relations receive additional loss weighting to counteract their lower positive frequency even after injection: attached_to (weight 1.8), adjacent_to (weight 1.5), on_top_of (weight 1.2), and under (weight 1.2). These weights multiply the ASL loss for positive examples of the respective relation, increasing the gradient signal for the most challenging relation classes.

Label smoothing is applied per relation to reduce overconfidence on ambiguous labels, particularly those derived from geometric rules where the boundary between positive and negative is not sharp. The smoothing values are: adjacent_to (0.12), in_front_of and behind (0.08 each), left_of and right_of (0.07 each), on_top_of and under (0.04 each), attached_to (0.05), higher_than and lower_than (0.04 each). Directional relations with perfectly deterministic rules receive the lowest smoothing, whilst adjacent_to — whose rule boundary is the most ambiguous — receives the highest.

3.9.3 Data Augmentation and Negative Subsampling

Each training scene is augmented with five additional variants: three rotations about the vertical axis (90°, 180°, 270°), two reflections (flip_x, flip_y), and a jitter augmentation that adds small random perturbations to all centroid positions. Scenes containing attached_to or adjacent_to relations — which are rare in 3DSSG — receive additional copies with extra flip_x and flip_y augmentations to oversample these rare-relation scenes. In total, each scene contributes up to seven graph instances to training (original plus six augmented), giving an augment_factor of 6. Augmentation is applied after the

train-validation split to prevent any augmented version of a validation scene from appearing in training.

Negative subsampling is applied per batch to manage the volume of negative examples. All positive edges are retained in every batch. Negative edges are randomly subsampled to achieve a target positive rate of approximately 30 per cent per batch. Soft labels with value ≥ 0.5 are treated as positive for the purposes of subsampling. This prevents the optimiser from spending the majority of each gradient step on trivially easy negative pairs.

3.9.4 Optimiser, Scheduler, and Warmup Configuration

The model is trained using the AdamW optimiser with weight decay 1e-4 and gradient clipping at max norm 2.0. The learning rate schedule consists of a linear warmup phase followed by cosine decay to zero. For GeoKAN-RBF, the initial learning rate is 2×10^{-3} with a 5-epoch warmup. For GeoKAN-Gamma, the initial learning rate is reduced to 5×10^{-4} with a 10-epoch warmup, reflecting the greater sensitivity of the fixed diagonal metric parameterisation to large early gradient steps. Training runs for a maximum of 100 epochs with early stopping at patience 15 — training halts if validation Macro F1 does not improve for 15 consecutive epochs. The batch size is 8 scenes. Positive class weights are computed per relation from the training set positive rate, capped at 8.0, and passed to the loss function to provide an additional signal for rare relations alongside the ASL focusing. GeoKAN-Gamma converges in approximately 1,093 seconds (18 minutes) on an NVIDIA RTX 3060; GeoKAN-RBF requires approximately 1,500 seconds due to the additional MetricNet forward pass.

3.9.5 Per-Relation Threshold Calibration

After training, each of the ten relation sigmoid outputs is thresholded independently using a threshold calibrated on the 3DSSG validation set. For each relation, the threshold that maximises the validation set F1 score for that relation is selected by sweeping candidate thresholds in [0.1, 0.9] at 0.05 intervals. This per-relation calibration is important because the marginal positive rates differ substantially across relations — higher_than and lower_than are positive for the majority of pairs, whilst attached_to is positive for very few — meaning a single global threshold cannot be optimal for all relations simultaneously. Hierarchical mutual exclusion constraints are applied after thresholding at evaluation time: for each axis-aligned relation pair (left_of / right_of, in_front_of / behind, higher_than / lower_than), if both relations are predicted for the same directed pair, the one with the lower sigmoid score is suppressed.

3.10 Symbolic Consistency Repair

3.10.1 Motivation and Design

Neural network predictions are not inherently constrained to satisfy logical consistency. A model that predicts spatial relations independently for each directed object pair may produce outputs that violate elementary spatial logic — for example, simultaneously

predicting higher_than(A, B) and higher_than(B, A), or predicting on_top_of(A, B) without the corresponding under(B, A). Such contradictions reduce the utility of the scene graph for downstream tasks that rely on logical inference, and they represent a form of prediction error that does not reflect genuine uncertainty but rather a failure of the model to enforce known structural constraints.

SceneGraphRepair is a zero-parameter post-processing module that enforces four categories of logical constraints on the predicted scene graph through deterministic fixed-point iteration. It carries no learnable parameters, requires no training data, and operates identically on any scene regardless of domain — making it fully domain-agnostic. Its design is motivated by the observation that many spatial constraints are universal physical facts rather than dataset-specific conventions: an object cannot simultaneously be both to the left of and to the right of another object; if A is higher than B then B must be lower than A.

3.10.2 Constraint Types

Four categories of constraints are enforced:

Inverse completeness. For each inverse relation pair — on_top_of ↔ under, higher_than ↔ lower_than, left_of ↔ right_of, in_front_of ↔ behind — if one direction is predicted with confidence above threshold, the inverse is added to the graph if not already present. This is the most impactful constraint: on the 3DSSG validation set it adds 13,027 inverse relations per 85 scenes (153 per scene on average), and on the tabletop cross-domain scenes it rescues the on_top_of relation by inferring it from the reliably predicted under relation (on_top_of F1 improving from 0.621 to 0.919 after repair).

Mutual exclusion. For each pair of mutually exclusive relations — on_top_of vs. under for the same directed pair, left_of vs. right_of, in_front_of vs. behind, higher_than vs. lower_than — if both are predicted, the one with the lower confidence score is removed. This eliminates direct directional contradictions where the model has predicted opposing relations simultaneously.

Asymmetry. For relations that are physically asymmetric — specifically higher_than and its directional analogues — if higher_than(A, B) and higher_than(B, A) are both predicted, the one with lower confidence is removed. This enforces the asymmetry property of strict ordering relations.

Transitivity. For transitive relations — higher_than and left_of chains — if higher_than(A, B) and higher_than(B, C) are both predicted, higher_than(A, C) is added if not already present. Transitivity inference is applied conservatively to avoid exponential chain propagation in large scenes.

3.10.3 Fixed-Point Iteration Algorithm

Constraint enforcement proceeds through alternating removal and addition passes until the predicted graph reaches a fixed point — a state in which no further constraint fires. In each iteration, the removal pass applies mutual exclusion and asymmetry constraints

to eliminate contradictions; the addition pass then applies inverse completeness and transitivity to infer new relations from the now-consistent graph. This alternation is necessary because adding an inverse relation in the addition pass may create a new contradiction that the next removal pass must resolve. In practice, convergence is fast: the algorithm requires an average of 3.3 iterations on the 3DSSG validation set and 1.9 iterations on the LERF scenes. The fixed-point is guaranteed to be reached because each iteration either removes at least one relation (strictly reducing the contradiction count) or adds relations without creating new contradictions.

The impact of symbolic repair varies by dataset. On the 3DSSG validation set, the model is already largely consistent due to the inverse consistency training loss, so repair resolves only 396 contradictions across 85 scenes (4.7 per scene) and produces a negligible change in Macro F1 (−0.002). On the LERF held-out scenes, repair adds 137 relations and improves mR@5 from 88.0 per cent to 92.9 per cent (+4.9 percentage points). On the tabletop cross-domain scenes, repair adds 88 relations entirely via inverse and symmetry inference and improves Micro F1 from 0.740 to 0.774 (+3.4 points), with the largest gains on on_top_of (+0.298 F1) and lower_than (+0.152 F1).

3.11 Evaluation Protocol

3.11.1 Metrics: Macro F1, Micro F1, and R@K

Three complementary metrics are used to evaluate scene graph prediction quality, each measuring a different aspect of performance.

Macro F1 is the unweighted average of per-relation F1 scores across all ten relation classes. For each relation, F1 = 2 × (precision × recall) / (precision + recall), computed using binary predictions obtained after per-relation threshold calibration. Macro F1 treats all relation classes equally regardless of their frequency, making it sensitive to performance on rare relations such as on_top_of and attached_to. It is the primary metric for measuring balanced classification performance across the full relation schema.

Micro F1 aggregates true positives, false positives, and false negatives across all relation classes before computing a single F1 score, effectively weighting each class by its frequency. Micro F1 is reported for cross-domain tabletop evaluation, where it provides a frequency-weighted view of overall prediction quality.

Predicate Recall at K (R@K) measures the fraction of ground truth relation triples for which the correct relation appears among the model's top-K highest-confidence predictions for that directed object pair, regardless of the threshold decision. R@3 and R@5 are reported throughout. R@K is a ranking metric rather than a classification metric — it evaluates whether the model correctly ranks the true relation highly, independently of whether the threshold produces the right binary decision. This makes R@K robust to threshold miscalibration and directly comparable across methods with different threshold strategies.

3.11.2 Baselines

Four baselines are used for comparison on the 3DSSG benchmark. ReLaGS (Arafa et al., 2026) and GaussianGraph (Li et al., 2025) are the two Gaussian-specific scene graph methods most directly comparable to this work, both operating on 3DGS representations but using large foundation models. ConceptGraphs (Gu et al., 2024) and Open3DSG (Koch et al., 2024) represent foundation-model-dependent approaches operating on RGB-D and point cloud inputs respectively. For the LERF evaluation, GaussianGraph is the primary baseline as it is the only prior method evaluated on the same four LERF scenes using the same positional relation query protocol. Results for all baselines are taken from their respective published papers; no reimplementation is performed.

3.11.3 Cross-Domain Evaluation Setup (Tabletop)

The eight tabletop evaluation scenes are used exclusively for zero-shot cross-domain evaluation — the GeoKAN-Gamma model trained on 3RScan is applied directly to tabletop scenes without any fine-tuning, threshold re-calibration, or scene-specific adaptation. The only legitimate adaptation applied is the suppression of the attached_to relation (setting predicted probabilities to zero), which is justified by the hard physical constraint that no objects in the tabletop scenes are wall-mounted or built-in — a domain expert constraint rather than a data-driven adaptation. The 3RScan-calibrated thresholds for all other nine relations are used unchanged. Ground truth for all eight scenes is constructed by build_gt_all_scenes.py, which runs the same deterministic HDBSCAN clustering pipeline as the evaluation script (RANSAC seed = 42), ensuring perfect coordinate alignment between ground truth centroids and predicted cluster centroids (Hungarian match distance = 0.000 for all eight scenes). Evaluation is conducted both without and with symbolic consistency repair to quantify the repair contribution independently of the base model.

3.11.4 Held-Out Evaluation Setup (LERF)

The four LERF scenes are used as a held-out evaluation set to assess in-domain generalisation to an unseen dataset at comparable scale to the training distribution. The evaluation protocol is identical to the 3DSSG validation protocol: the GeoKAN-Gamma model with 3RScan-calibrated thresholds is applied without modification, and R@1, R@3, and R@5 are computed against geometry-derived ground truth. Results are reported both before and after symbolic consistency repair. Comparison against GaussianGraph is conducted using the mR@1, mR@3, and mR@5 metrics on positional relation queries, matching the evaluation protocol reported in Table V of Li et al. (2025).

4.Results

4.1 In-Domain Performance on 3DSSG Validation Set

4.1.1 Model Evolution Table

Table 4.1 presents the full evolution of the GeoKANRelationGNN architecture from the initial GAT baseline through to the final GeoKAN-Gamma model. Each row corresponds

to a distinct model version with a specific architectural or training change, trained and evaluated on the 3RScan / 3DSSG dataset unless otherwise noted.

Version	Key Change	Data	Macro F1
v1–v3	Single-head GAT, 8-dim node features, softmax loss	ScanNet (geo-derived)	0.35 (unstable)
v4	Multi-label BCE, 10-dim node features, signed delta edge features	ScanNet (geo-derived)	0.42
v5	Fixed data leakage (augmentation after split), all-pairs negatives	ScanNet (geo-derived)	0.44
v6	14-dim edge features with contact signals (vert_gap, contact_score, support_overlap)	ScanNet (geo-derived)	0.49
v7	17-dim edge features, dual-head GAT, asymmetric focal loss, threshold tuning	ScanNet (geo-derived)	0.52
GeoKAN-v1	GeoKAN heads replacing MLP, 3RScan real Gaussian data, 12-class schema	3RScan / 3DSSG	0.505
GeoKAN-v2	22-dim scale-invariant edge features (avg_diag normalisation)	3RScan / 3DSSG	0.48
GeoKAN-v3	Incorrect ASL configuration ($\gamma^{+} = 4.0$, prob_margin = 0.05)	3RScan / 3DSSG	0.46
GeoKAN-RBF	Rule injection, correct ASL ($\gamma^{-} = 4.0$, $\gamma^{+} = 0.0$), 10-class schema	3RScan / 3DSSG + injected	0.919
GeoKAN-Wavelet	Mexican hat wavelet basis replacing RBF	3RScan / 3DSSG + injected	0.914
GeoKAN-Gamma	Fixed diagonal metric, LR = 5×10^{-4}, warmup = 10 epochs	3RScan / 3DSSG + injected	0.926

Table 4.1: Model evolution from GAT baseline to GeoKAN-Gamma. The jump from v7 (0.52) to GeoKAN-RBF (0.919) is attributable primarily to rule-based label injection and correct ASL configuration.

4.1.2 GeoKAN-Gamma Final Results

GeoKAN-Gamma achieves Macro F1 = 0.926, R@3 = 0.878, and R@5 = 0.983 on the 3DSSG validation set of 85 held-out scenes containing 323,237 ground truth relation triples. The model contains 1,018,414 parameters and completes inference in under two seconds on an NVIDIA RTX 3060 GPU using less than 1.5 GB of GPU memory.

4.1.3 Per-Relation F1 Breakdown

Table 4.2 presents the per-relation F1 scores for all three GeoKAN variants on the 3DSSG validation set.

Relation	GeoKAN-Gamma	GeoKAN-RBF	GeoKAN-Wavelet
on_top_of	0.846	0.839	0.833
under	0.857	0.844	0.826
attached_to	0.940	0.936	0.934
adjacent_to	0.945	0.942	0.943
left_of	0.957	0.955	0.950
right_of	0.956	0.955	0.950
in_front_of	0.902	0.888	0.875
behind	0.901	0.889	0.884
higher_than	0.977	0.968	0.968
lower_than	0.976	0.977	0.972
Macro F1	0.926	0.919	0.914

Table 4.2: Per-relation F1 scores for all three GeoKAN variants. GeoKAN-Gamma's largest gains over RBF are on in_front_of (+1.4 pp), behind (+1.2 pp), and under (+1.3 pp).

4.1.4 Training Convergence

GeoKAN-RBF converges rapidly from the first epoch, reaching Macro F1 = 0.711 at epoch 1, 0.816 at epoch 5, 0.903 at epoch 10, and the final value of 0.919 by epoch 15. This fast convergence reflects the strong geometric signal provided by the injected labels — the model learns the directional relations almost immediately and progressively refines the contact relation boundaries over subsequent epochs. GeoKAN-Gamma follows a similar trajectory with a slower initial ramp due to the longer 10-epoch warmup, but reaches a higher final value of 0.926 with greater training stability. Total training time is approximately 18 minutes for GeoKAN-Gamma and 25 minutes for GeoKAN-RBF on an NVIDIA RTX 3060.

4.2 GeoKAN Basis Function Ablation

4.2.1 GeoKAN-Gamma vs. RBF vs. Wavelet

Table 4.3 presents a direct comparison of the three GeoKAN basis function variants under identical training conditions — the same dataset, augmentation strategy, loss configuration, and evaluation protocol — differing only in the basis function and metric parameterisation of the GeoKAN layers.

Variant	Metric Type	Basis Function	Macro F1	R@3	R@5	Params	Train Time (s)
GeoKAN-Gamma	Fixed diagonal: g_i = softplus(γ_i)	— (no RBF; warped linear)	0.926	0.878	0.983	1,018,414	1,093
GeoKAN-RBF	Adaptive: MetricNet MLP per sample	RBF: $\exp(-\gamma(z-c_k)^2)$, K=12	0.919	0.875	0.982	1,071,918	1,500
GeoKAN-Wavelet	Adaptive: MetricNet MLP per sample	Mexican hat: $(1-t^2)\exp(-t^2/2)$	0.914	0.872	0.979	1,071,918	1,268

Table 4.3: GeoKAN basis function ablation. All variants trained on 3RScan / 3DSSG with identical configuration. GeoKAN-Gamma is the best-performing variant despite having the simplest metric parameterisation.

Three findings emerge from this ablation.

GeoKAN-Gamma outperforms the adaptive variants. The fixed diagonal metric, which learns a single scalar scaling per input dimension with no dependence on the input value, achieves the highest Macro F1 (0.926) and R@5 (0.983) despite being the architecturally simplest variant. It also has 53,504 fewer parameters than the RBF and Wavelet variants, as the MetricNet MLP is replaced by a single learnable vector γ. This result indicates that for the spatial relation prediction task, the additional flexibility of an input-dependent adaptive metric does not translate into improved predictive performance — the geometric features are sufficiently well-structured that a fixed re-scaling of each dimension captures the relevant variation without requiring per-sample adaptation.

GeoKAN-Gamma required different training hyperparameters. The fixed diagonal metric is more sensitive to large gradient steps in early training, as the metric parameters γ interact directly with the basis expansion without the stabilising effect of the MetricNet's nonlinear path. A lower learning rate (5×10^{-4} vs. 2×10^{-3}) and a longer warmup (10 vs. 5 epochs) were necessary for stable training. The gains from GeoKAN-Gamma are concentrated on the relations that benefit most from consistent axis-aligned scaling: in_front_of (+1.4 pp over RBF), behind (+1.2 pp), and under (+1.3 pp) — all relations whose ground truth is strongly aligned with a single feature axis.

GeoKAN-Wavelet performs worst. The Mexican hat wavelet basis, whilst theoretically appealing for its ability to represent oscillatory patterns, produces the lowest Macro F1 (0.914) across all three variants. The negative lobes of the wavelet function — which produce negative activations for inputs away from the centre — cause destructive interference in the multi-label classification setting, where the model must simultaneously activate multiple output heads for a given input. The RBF basis, with its non-negative activations, does not suffer from this interference effect and outperforms the wavelet variant across all metrics.

4.3 Comparison with State-of-the-Art

4.3.1 vs. ReLaGS (Arafa et al., CVPR 2026)

ReLaGS is the current state-of-the-art Gaussian-based scene graph method and the most directly comparable system to this work. It constructs a hierarchical language-distilled Gaussian scene representation using CLIP, SAM, and a Jina embedding model, then applies a graph neural network to predict spatial and semantic relations from the resulting language-augmented embeddings. On the 3DSSG benchmark, ReLaGS reports R@3 = 0.790 and R@5 = 0.870. GeoKAN-Gamma surpasses these figures by 8.8 percentage points on R@3 (0.878 vs. 0.790) and 11.3 percentage points on R@5 (0.983 vs. 0.870), while using approximately 0.07 per cent of the parameter count (1.018M vs. ~1.5B) and completing inference in under two seconds rather than 12.6 minutes. This comparison demonstrates that a geometry-first approach, trained on structured geometric features with rule-augmented labels, can substantially outperform a state-of-the-art foundation-model-dependent system on the established benchmark for this task.

4.3.2 vs. ConceptGraphs (Gu et al., ICRA 2024)

ConceptGraphs constructs open-vocabulary 3D scene graphs by fusing outputs from 2D foundation models — including class-agnostic instance segmentation, CLIP feature extraction, and large language model relation inference — into a persistent 3D object map. On the 3DSSG benchmark it achieves R@3 = 0.740 and R@5 = 0.790. GeoKAN-Gamma exceeds ConceptGraphs by 13.8 percentage points on R@3 and 19.3 percentage points on R@5, whilst the geometry-first approach requires no language model, no segmentation model, and no CLIP encoder at inference time. The ConceptGraphs result is noteworthy because it reflects a fundamentally different paradigm — open-vocabulary, LLM-reasoned relations — yet is surpassed by a model that predicts relations purely from 3D geometry.

4.3.3 vs. Open3DSG (Koch et al., CVPR 2024)

Open3DSG is the first open-vocabulary 3D scene graph method operating on point cloud inputs, achieving R@3 = 0.580 and R@5 = 0.650 on the 3DSSG benchmark using CLIP co-embedding. GeoKAN-Gamma exceeds Open3DSG by 29.8 percentage points on R@3 and 33.3 percentage points on R@5. Open3DSG operates on point clouds rather than Gaussian splats, meaning it does not exploit the per-Gaussian covariance and opacity information available in the 3DGS representation. This comparison suggests that the richer geometric signal of Gaussian splats, combined with geometry-first feature engineering, provides a substantial representational advantage over point cloud methods even when those methods are augmented with large vision-language models.

4.3.4 Computational Efficiency Comparison

Table 4.4 summarises the performance and computational footprint of all compared methods.

Method	R@3	R@5	Foundation Models	Params	Inference Time	GPU Memory
GeoKAN-Gamma (ours)	0.878	0.983	None	1.018M	< 2 sec	< 1.5 GB
GeoKAN-RBF (ours)	0.875	0.982	None	1.072M	< 2 sec	< 1.5 GB
GeoKAN-Wavelet (ours)	0.872	0.979	None	1.072M	< 2 sec	< 1.5 GB
ReLaGS	0.790	0.870	CLIP + SAM + Jina	~1.5B	12.6 min	7.5 GB
ConceptGraphs	0.740	0.790	LLM + CLIP	~1B+	—	—
Open3DSG	0.580	0.650	CLIP	~400M	—	—

Table 4.4: Comparison with state-of-the-art methods on the 3DSSG benchmark. GeoKAN-Gamma achieves the highest R@3 and R@5 whilst using the fewest parameters, the shortest inference time, and the lowest GPU memory footprint of any compared method.

The efficiency advantage of GeoKAN-Gamma over ReLaGS is particularly striking: GeoKAN-Gamma uses 0.07 per cent of ReLaGS's parameter count, completes inference approximately 380 times faster, and requires 5 times less GPU memory — whilst outperforming it on both benchmark metrics. This positions the geometry-first approach as not merely competitive but dominant in the efficiency-performance trade-off space for Gaussian-based scene graph generation.

4.4 Impact of Symbolic Consistency Repair

4.4.1 On the 3DSSG Validation Set

On the 3DSSG validation set of 85 scenes, symbolic consistency repair has a near-neutral impact on Macro F1, reducing it by 0.002 points (from 0.928 to 0.926). This near-zero impact is expected and reflects a positive property of the trained model: GeoKAN-Gamma, having been trained with the inverse consistency loss L_inv, already produces largely consistent predictions on in-domain data. Across 85 validation scenes, repair resolves only 396 contradictions (4.7 per scene on average) — an extremely low contradiction rate given that each scene contains hundreds of directed object pairs. The repair module does add 13,027 inverse relations (153 per scene on average) via inverse completion, but because the model's threshold-calibrated predictions already capture most of these inverse relations on the in-domain validation set, the additional inverses do not substantially change the F1 calculation. The negligible F1 change on 3DSSG confirms that symbolic repair does not degrade in-domain performance whilst providing a safety guarantee of logical consistency.

4.4.2 On the LERF Scenes

The impact of symbolic repair is substantially larger on the four LERF held-out scenes, where the model operates on an unseen dataset without threshold re-calibration. Table 4.5 presents per-scene results before and after repair.

Scene	Objects	GT Triples	mR@5 Before	mR@5 After	Relations Added
ramen	13	640	0.898	0.948	—
teatime	10	372	0.922	0.941	—
waldo_kitchen	10	368	0.859	0.921	—
figurines	10	390	0.831	0.892	—
Aggregate	—	1,770	0.880	0.929	137

Table 4.5: Impact of symbolic consistency repair on the four LERF evaluation scenes. Repair adds 137 relations via inverse completion and improves aggregate mR@5 by 4.9 percentage points.

Repair adds 137 relations across the four scenes (34.25 per scene on average) via the inverse completion mechanism, with 2 contradictions resolved. The mR@5 improvement of 4.9 percentage points — from 0.880 to 0.929 — arises because on unseen scenes the model's threshold-calibrated predictions are less complete than on the in-domain validation set: the model correctly ranks the true relation highly (hence R@5 before repair is already 88.0 per cent) but does not always fire the threshold for both a relation and its inverse. The repair module infers the missing inverse from the confidently predicted direction, recovering these missed relations without any retraining or threshold adjustment.

4.5 Held-Out Scene Evaluation: LERF (In-Domain Generalisation)

4.5.1 Per-Scene Results

The four LERF scenes — ramen, teatime, waldo_kitchen, and figurines — serve as a held-out evaluation set for assessing in-domain generalisation to an unseen dataset. These scenes cover indoor environments at 1–5 metre scale, placing them within the same scale

class as the 3RScan training distribution (3–8 metres), and are therefore a test of cross-dataset rather than cross-domain generalisation. The GeoKAN-Gamma model with 3RScan-calibrated thresholds is applied without modification. Table 4.6 presents per-scene results after symbolic consistency repair.

Scene	Objects	GT Triples	mR@1	mR@3	mR@5
ramen	13	640	0.891	0.897	0.948
teatime	10	372	0.925	0.925	0.941
waldo_kitchen	10	368	0.856	0.872	0.921
figurines	10	390	0.808	0.826	0.892
Aggregate	—	1,770	0.872	0.882	0.929

Table 4.6: Per-scene LERF results for GeoKAN-Gamma after symbolic consistency repair.

Performance is consistently strong across all four scenes. The teatime scene achieves the highest mR@1 (0.925), reflecting its relatively simple spatial arrangement with clear vertical and horizontal separations between objects. The figurines scene achieves the lowest mR@5 (0.892), which is consistent with its dense arrangement of small decorative objects at similar heights — a configuration that challenges the higher_than and lower_than predictions where Z differences are small. The ramen scene, despite having the largest object count (13 objects, yielding 156 directed pairs), achieves a strong mR@5 of 0.948, demonstrating that the model scales well to scenes with higher object counts.

4.5.2 Comparison vs. GaussianGraph on LERF

Table 4.7 compares GeoKAN-Gamma against GaussianGraph (Li et al., 2025) on the four LERF scenes using the positional relation query protocol from Table V of Li et al. (2025).

Method	Foundation Models	Params	mR@1	mR@3	mR@5
GeoKAN-Gamma + Repair (ours)	None	1.018M	0.872	0.882	0.929
GeoKAN-Gamma (before repair)	None	1.018M	0.235	0.653	0.880
GaussianGraph LLaVA-1.6 + 3D correction	CLIP+SAM+LLaVA-1.6+GroundingDINO	~2B+	0.568	0.613	0.632
GaussianGraph LLaVA-1.6, no correction	CLIP+SAM+LLaVA-1.6+GroundingDINO	~2B+	0.407	0.450	0.498

Method	Foundation Models	Params	mR@1	mR@3	mR@5
GaussianGraph BLIP2 + 3D correction	CLIP+SAM+BLIP2	—	0.355	0.379	0.401

Table 4.7: Comparison against GaussianGraph on the four LERF scenes using positional relation queries.

GeoKAN-Gamma with symbolic repair achieves mR@5 = 0.929, exceeding GaussianGraph's best result of 0.632 (LLaVA-1.6 with 3D correction) by 29.7 percentage points. This result is particularly significant because GaussianGraph was specifically designed and evaluated on these LERF scenes, whilst GeoKAN-Gamma was trained exclusively on 3RScan with no LERF-specific adaptation. The comparison also highlights the role of symbolic repair: without it, mR@1 falls to 0.235, indicating that the raw model predictions rank many correct relations outside the top position. Repair's inverse completion mechanism recovers these by inferring missing inverse relations, raising mR@1 from 0.235 to 0.872 — a 63.7 percentage point improvement on the rank-1 metric alone.

It is important to note that the LERF evaluation does not constitute cross-domain generalisation. The 1–5 metre scale of the LERF scenes is comparable to the 3–8 metre scale of the 3RScan training scenes; both fall within the same indoor room-scale class. The strong LERF performance therefore reflects in-domain generalisation to an unseen dataset rather than adaptation to a different scale or scene type, a distinction that is examined further in the cross-domain tabletop evaluation in Section 4.6.

4.6 Cross-Domain Evaluation: Custom Tabletop Scenes

4.6.1 Ground Truth Construction Methodology

Eight custom tabletop scenes (scene_06 through scene_13) were captured and reconstructed for the sole purpose of cross-domain evaluation. Each scene was captured using a smartphone camera at approximately 60 frames, reconstructed via NerfStudio splatfacto, and evaluated using the same GeoKAN-Gamma model trained on 3RScan — with no scene-specific fine-tuning, threshold adjustment, or model adaptation.

Ground truth relation files were constructed programmatically using build_gt_all_scenes.py rather than manual annotation. The script runs the same deterministic HDBSCAN clustering pipeline as the evaluation code (RANSAC seed = 42), reads the actual cluster centroids produced by that clustering, and derives all pairwise spatial relations geometrically from those centroids. Cluster-to-object assignments were verified by visual inspection of centroid positions and point counts. This approach ensures perfect coordinate alignment between ground truth and evaluation: the Hungarian match distance between predicted and ground truth cluster centroids is 0.000 for all eight scenes, eliminating any possibility of mismatch between the object instances used to construct ground truth and those used during prediction.

Three scenes (scene_09, scene_12, scene_13) exhibit a Z-DOWN coordinate convention where more negative Z values correspond to physically higher positions; this is handled during ground truth construction by negating Z before deriving directional relations for those scenes. Five scenes contain a verified stacking configuration (router placed on top of the Agaro box): scenes 06, 08, 10, 12, and 13.

4.6.2 Zero-Shot Aggregate Results

Table 4.8 presents the aggregate zero-shot cross-domain results across all eight tabletop scenes, reported both without and with symbolic consistency repair.

Setting	Micro F1	Precision	Recall	R@3	R@5
Zero-shot, no repair	0.740	0.815	0.677	0.758	0.927
Zero-shot + symbolic repair	0.774	0.773	0.776	0.758	0.927

Table 4.8: Aggregate zero-shot cross-domain results on 8 tabletop scenes. R@K is unchanged by repair as expected — the ranking metric is not affected by inverse completion.

The zero-shot results demonstrate meaningful generalisation despite the substantial scale gap between training (3–8 m room-scale) and evaluation (0.3–0.8 m tabletop). R@5 = 0.927 indicates that the correct relation appears in the model's top-5 predictions for 92.7 per cent of ground truth triples, and R@3 = 0.758 shows that it ranks in the top 3 for 75.8 per cent. The Micro F1 of 0.740 (without repair) reflects a high-precision, lower-recall profile (precision = 0.815, recall = 0.677), consistent with the model being conservative at the 3RScan-calibrated thresholds on the smaller-scale tabletop scenes.

4.6.3 Per-Relation Analysis

Table 4.9 presents per-relation R@3 and R@5 results, revealing which spatial relations transfer well across the scale domain gap and which degrade.

Relation	R@3	R@5	Transfer Quality
adjacent_to	0.981	1.000	Excellent — proximity is scale-invariant
left_of	0.942	1.000	Excellent — X-axis ordering is scale-invariant
higher_than	0.882	0.953	Good
right_of	0.750	0.981	Good — slight asymmetry vs. left_of
on_top_of	0.684	0.895	Moderate — table context helps
lower_than	0.659	0.729	Weak — small Z differences on tabletop
in_front_of	0.538	1.000	Poor R@3, perfect R@5 — consistent rank-4/5 prediction

Relation	R@3	R@5	Transfer Quality
behind	0.481	0.981	Poor R@3, near-perfect R@5 — same pattern

Table 4.9: Per-relation cross-domain transfer on 8 tabletop scenes. Relations based on X-axis and proximity transfer best; Y-axis (depth) relations and lower_than are weakest.

Relations that depend on the X-axis (left_of, right_of) and on proximity (adjacent_to) transfer perfectly or near-perfectly, as these geometric relationships are inherently scale-invariant. The in_front_of and behind relations show a striking pattern: R@3 is poor (0.538 and 0.481) but R@5 is near-perfect (1.000 and 0.981). This indicates the model consistently ranks these relations 4th or 5th rather than in the top 3, reflecting ambiguity in the Y-axis (depth) dimension on tabletop scenes where objects are densely packed with small Y separations compared to the large room-scale depths seen during training. The lower_than relation is the weakest performer overall (R@5 = 0.729), as all tabletop objects are at similar heights within the narrow vertical range of the tabletop surface, making the small Z differences fall below the learned detection threshold calibrated on room-scale scenes where height differences are much larger.

4.6.4 Per-Scene Results

Table 4.10 presents per-scene R@3 and R@5 results across the eight tabletop evaluation scenes.

Scene	Z-Convention	Stacking	R@3	R@5
scene_06	Z-UP	router on agaro_box	0.750	0.955
scene_07	Z-UP	None	0.776	0.931
scene_08	Z-UP	router on agaro_box	0.733	0.883
scene_09	Z-DOWN	None	0.793	0.966
scene_10	Z-UP	router on agaro_box	0.804	0.911
scene_11	Z-UP	None	0.790	0.984
scene_12	Z-DOWN	router on agaro_box	0.733	0.917
scene_13	Z-DOWN	router on agaro_box	0.697	0.864
Mean	—	—	0.760	0.926

Table 4.10: Per-scene cross-domain results on 8 tabletop evaluation scenes.

Performance is consistent across scenes, with R@3 ranging from 0.697 to 0.804 and R@5 from 0.864 to 0.984. Scene_13 achieves the lowest R@3 (0.697), which coincides with both a Z-DOWN convention and a stacking configuration — two complicating factors simultaneously present. Scene_11 achieves the highest R@5 (0.984), reflecting its straightforward side-by-side arrangement without stacking. No systematic performance difference is observed between Z-UP and Z-DOWN scenes, confirming that the Z-axis correction applied during clustering correctly normalises the coordinate convention before feature computation.

4.6.5 Impact of Symbolic Repair Cross-Domain

Table 4.11 presents the per-relation impact of symbolic repair on the tabletop cross-domain scenes.

Relation	F1 Before Repair	F1 After Repair	Δ F1	Mechanism
on_top_of	0.621	0.919	+0.298	under reliably predicted (F1=0.932); repair infers on_top_of as inverse
lower_than	0.768	0.920	+0.152	higher_than well predicted; repair adds 32 missing lower_than counterparts
behind	0.308	0.429	+0.121	in_front_of inverses added
right_of	0.800	0.860	+0.060	left_of inverses added
adjacent_to	0.675	0.568	−0.107	Minor over-prediction artifact from bidirectional inverse firing

Table 4.11: Per-relation impact of symbolic repair on tabletop cross-domain scenes. The largest gain is on on_top_of (+0.298), rescued entirely by inverse completion from under.

The most striking result is the on_top_of recovery: the model predicts under with F1 = 0.932 cross-domain, but on_top_of with only F1 = 0.621 before repair. The asymmetry arises because the under relation is represented by edge features computed from the object-below's perspective — features that transfer well from 3RScan — whilst on_top_of features computed from the object-above's perspective are less robust at tabletop scale. Symbolic repair corrects this asymmetry entirely through inverse completion, raising on_top_of F1 to 0.919. The one negative impact — adjacent_to dropping by 0.107 — is a minor artifact where the bidirectional inverse firing slightly over-predicts adjacency. Symbolic repair provides +3.4 Micro F1 points cross-domain with zero parameters and no retraining, confirming its value as a domain-agnostic correction mechanism.

5.DISCUSSION

5.1 Is Geometry Sufficient? The Case Against Foundation Models

The central claim of this project is that spatial relations between objects in a 3D scene can be predicted accurately from geometric features alone, without any visual language model, segmentation model, or semantic label at inference time. The results reported in

Chapter 4 provide strong empirical support for this claim. GeoKAN-Gamma achieves Macro F1 = 0.926 and R@5 = 0.983 on the 3DSSG benchmark — surpassing every compared method including those that use foundation models with hundreds of millions to over a billion parameters.

This outcome is perhaps less surprising than it initially appears. The ten spatial relations in the project's schema are, by definition, geometric predicates. Left_of is a statement about X-axis ordering. Higher_than is a statement about Z-axis ordering. On_top_of is a statement about vertical proximity and footprint overlap. None of these relations require knowing that an object is a chair, a lamp, or a bottle. The geometric facts that define them are fully derivable from centroid positions and bounding volumes — precisely the information encoded in the 10-dimensional node features and 22-dimensional edge features used here. The question was never whether geometry contains this information; it clearly does. The question was whether a model could be trained to extract it reliably, given the sparsity of available annotations. Rule-based label injection resolves that bottleneck, and GeoKAN provides the architectural flexibility to learn the non-linear geometric decision boundaries that separate, for example, on_top_of from merely higher_than.

The comparison with foundation-model-based methods reveals a structural inefficiency in the current dominant paradigm. Methods such as ReLaGS and GaussianGraph use vision-language models primarily to identify what objects are — to assign semantic labels that are then used to reason about how objects relate. But the relations being predicted are geometric, not semantic. A model does not need to know that object A is a router and object B is a box to determine that A is resting on B; it needs to know that A's bottom face is close to B's top face and that A's footprint overlaps B's footprint. Using a language model to derive this geometric fact adds approximately 1.5 billion parameters, 12 minutes of inference time, and 7.5 GB of memory to a computation that a 1-million-parameter geometry-only model performs in under two seconds with higher accuracy. This mismatch suggests that the field has been importing foundation model infrastructure into a problem that does not require it, at substantial computational cost.

This is not an argument against foundation models in scene understanding broadly. Open-vocabulary object recognition, natural language scene querying, and semantic relation prediction — where the relation type itself is not geometric — genuinely benefit from language grounding. But for the specific task of predicting a fixed vocabulary of geometric spatial relations, the results here suggest that geometry alone, when properly exploited, is not merely sufficient but superior.

5.2 Why GeoKAN-Gamma Outperforms the Adaptive RBF Variant

The ablation study in Section 4.2 produces a counterintuitive result: the architecturally simplest GeoKAN variant — a fixed diagonal metric with no input-dependent adaptation — outperforms both the adaptive RBF and wavelet variants that use a MetricNet MLP to compute per-sample metrics. This outcome merits careful examination, as it runs

counter to the general expectation that more expressive models should perform better given sufficient data.

The key insight lies in the nature of the input features. The 10-dimensional node features and 22-dimensional edge features used in this project are already highly engineered: they are scale-normalised, clipped, and designed to encode specific geometric signals with minimal redundancy. The delta_x feature directly encodes left-right displacement; the vert_gap feature directly encodes vertical contact proximity; the z_rank feature encodes ordinal height. These features are not raw sensor measurements requiring complex nonlinear transformation — they are near-sufficient statistics for the relations they are designed to predict. In this setting, the primary function of the geometric metric is not to discover a complex nonlinear warping of the feature space, but to learn which dimensions are most informative for each prediction head and scale them appropriately. A fixed diagonal metric — which learns exactly this: one scale factor per dimension — is precisely what the task requires. The MetricNet MLP in GeoKAN-RBF adds input-dependent adaptation, but since the features are already well-calibrated, this additional flexibility fits noise rather than signal, resulting in marginally lower generalisation performance.

A second factor is training stability. The fixed diagonal metric in GeoKAN-Gamma interacts directly with the RBF centres during basis expansion, meaning that large gradient steps in early training can displace the effective input distribution significantly. This sensitivity required a lower learning rate (5×10^{-4} vs. 2×10^{-3}) and longer warmup (10 vs. 5 epochs) for stable convergence. Once these hyperparameters were identified, GeoKAN-Gamma trained more stably than GeoKAN-RBF and converged to a higher final value. The MetricNet in GeoKAN-RBF introduces a nonlinear buffer between the raw features and the metric output, which moderates gradient flow and allows training at a higher learning rate — but at the cost of the metric becoming input-dependent in ways that introduce variance across samples.

The GeoKAN-Wavelet result provides a complementary insight. The Mexican hat wavelet basis produces negative activations for inputs far from each centre, which in a multi-label classification setting causes destructive interference: the same feature value that activates one relation's output may simultaneously suppress another's through negative wavelet lobes. The RBF basis, with its strictly non-negative activations, avoids this pathology. The consistent ranking Gamma > RBF > Wavelet across all ten relations and all three evaluation sets suggests that the ordering reflects a genuine architectural property rather than a random fluctuation.

The practical implication is clear: for tasks involving well-engineered geometric features with known physical interpretations, a simpler metric parameterisation with carefully tuned training dynamics outperforms a more expressive adaptive variant. Complexity should be introduced only where the data genuinely requires it.

5.3 The Role of Rule-Based Label Injection

The model evolution table in Section 4.1.1 reveals that the single largest performance jump in the entire development history occurs between GeoKAN-v3 (Macro F1 = 0.46) and GeoKAN-RBF (Macro F1 = 0.919) — a gain of 0.459 F1 points. This jump is not attributable to any architectural change: the GeoKAN layer was already present in GeoKAN-v1. It is attributable primarily to rule-based label injection combined with the correction of the ASL configuration. Disentangling these two contributions precisely is not straightforward, but the trajectory from v7 (0.52, standard model with correct ASL but no injection and no GeoKAN) to GeoKAN-v1 (0.505, GeoKAN architecture but no injection) to GeoKAN-RBF (0.919, GeoKAN with injection) makes clear that injection is the dominant driver.

The mechanism is straightforward. Before injection, the 3DSSG training set contains approximately 40,000 positive examples each for left_of and right_of — the human-annotated subset. After injection, these counts rise to 450,434 each. The model trained without injection sees the vast majority of geometrically valid left_of pairs labelled as negative — not because they are not left_of, but because the annotator did not choose to label them. This false negative signal directly suppresses recall for these relations. After injection, the model receives a training signal that accurately reflects the geometric reality: if A's centroid X is less than B's centroid X, that pair is positive for left_of, regardless of whether a human annotator happened to label it.

It is important to be precise about what injection does and does not do. It does not introduce any information that is not already present in the data — the centroid positions from which directional labels are derived are the same positions used at inference time. It does not create a new capability; it removes an artificial suppression that the annotation process had imposed on an existing capability. The model always had access to the geometric signal for left_of; injection simply corrects the false negative labels that were preventing it from learning to use that signal.

The lower confidence values assigned to contact relation injections (0.55–0.75) reflect a genuine epistemological distinction. Directional relations are geometric facts with confidence 1.0: if A's centroid is to the left of B's centroid, the rule is correct with certainty. Contact relations such as on_top_of depend on vertical gap thresholds and footprint overlap criteria that are approximations of a continuous physical reality — an object near but not quite touching a surface may or may not be on_top_of depending on the annotator's interpretation. Assigning lower confidence to these injected labels is not conservatism but accuracy: it reflects the genuine uncertainty of the rule.

The annotation sparsity problem identified by Wu et al. (2020) in 2D scene graph datasets is therefore even more acute in 3D: the 3DSSG annotation rate of 80–120 pairs per scene out of ~870 possible pairs (roughly 10–14 per cent coverage) means that any model trained without addressing this sparsity is learning from a dataset that is 86–90 per cent false negatives by construction. Rule-based injection is not a data augmentation trick; it is a correction of a fundamental flaw in the training signal.

5.4 Symbolic Repair: Gains, Trade-offs, and the Inverse Completion Mechanism

Symbolic consistency repair occupies an unusual position in the LogicSplat pipeline: it has zero learnable parameters, requires no training data, and yet produces measurable improvements in scene graph quality — particularly in cross-domain settings where the base model's threshold-calibrated predictions are imperfect. Understanding precisely why repair helps, and where it does not, requires examining the mechanism rather than the aggregate numbers.

The dominant mechanism across all three evaluation settings is inverse completion rather than contradiction resolution. On the 3DSSG validation set, repair resolves only 396 contradictions across 85 scenes but adds 13,027 inverse relations. On the LERF scenes, repair resolves 2 contradictions but adds 137 relations. On the tabletop scenes, repair resolves 0 contradictions but adds 88 relations. This pattern reveals an important property of GeoKAN-Gamma: it is already largely self-consistent — the inverse consistency training loss L_inv successfully suppresses logical contradictions during training, so few contradictions remain at test time. The value of repair lies almost entirely in completing missing inverse relations rather than removing impossible ones.

The tabletop on_top_of recovery illustrates the mechanism precisely. The model predicts under with F1 = 0.932 cross-domain, whilst predicting on_top_of with only F1 = 0.621. This asymmetry is not random — it reflects a systematic difference in how the two relations are represented in the edge feature space. The edge features for under(B, A) are computed from B's perspective, encoding how far below A's bottom face B's top face sits. These features transfer well across the scale domain gap because the relative vertical positioning of objects is captured by the normalised vert_gap and contact_score features. The on_top_of(A, B) features, computed from A's perspective, are less robust because A is the smaller tabletop object whose position relative to the table surface involves Z differences that fall below the room-scale thresholds. Repair bridges this asymmetry by treating the reliable under prediction as evidence for the inverse on_top_of, recovering 0.298 F1 points without any model change.

The one negative impact of repair — adjacent_to dropping by 0.107 F1 on the tabletop scenes — is informative. Adjacent_to is symmetric: if A is adjacent to B, then B is adjacent to A. When the model predicts adjacent_to(A, B) with sufficient confidence, repair adds adjacent_to(B, A) as its inverse. On the tabletop scenes, where objects are densely packed and many pairs are near the adjacency distance threshold, this symmetric completion slightly over-predicts adjacency for pairs that are geometrically marginal. This suggests a refinement: symmetric relation completion should apply a confidence penalty — adding the inverse only if the original prediction exceeds a higher threshold — to avoid over-firing on borderline cases. This is left as a direction for future work.

The broader design principle is that symbolic repair is most valuable when a model's prediction errors are systematic rather than random — when it consistently predicts one direction of a relation reliably and the other direction poorly. Random prediction errors cannot be corrected by logic alone; systematic asymmetries can. GeoKAN-Gamma exhibits exactly this property in the cross-domain setting, making symbolic repair a

particularly effective complement to the learned model rather than a generic post-processing step.

5.5 Cross-Domain Behaviour: What Transfers and What Does Not

The per-relation tabletop results in Section 4.6.3 reveal a clear and interpretable pattern of cross-domain transfer that maps directly onto the geometric properties of each relation.

Relations that transfer perfectly or near-perfectly — adjacent_to (R@5 = 1.000), left_of (R@5 = 1.000), right_of (R@5 = 0.981) — share a common property: they are defined by a single geometric dimension that is inherently scale-invariant when expressed as a ratio or rank. Left_of is simply a statement about which object has the smaller X centroid; this comparison holds regardless of whether the absolute X separation is 2 metres (room scale) or 5 centimetres (tabletop scale). Adjacent_to is defined by a threshold expressed as a multiple of the average object diagonal — a relative measure that automatically adjusts to the scene scale. These relations require no calibrated absolute threshold; they are rank or ratio comparisons that the scale-invariant normalisation strategy preserves exactly across domain boundaries.

Relations that degrade moderately — on_top_of (R@5 = 0.895), higher_than (R@5 = 0.953) — involve vertical relationships that are directionally correct but rely on magnitude comparisons. Higher_than transfers well because Z-rank ordering is preserved across scales; the ordinal rank of object heights is the same whether objects are separated by 0.5 metres or 5 centimetres. On_top_of is more sensitive because it depends on the vert_gap and contact_score features, which involve absolute vertical gap thresholds relative to avg_diag. When avg_diag shrinks from ~1 m at room scale to ~0.15 m at tabletop scale, a normalised gap value that indicates contact in a room becomes a normalised gap value that indicates hovering on a tabletop. The contact threshold calibrated on 3RScan is slightly too loose for tabletop geometry, causing some misclassification before symbolic repair corrects it via inverse completion.

The most revealing failure is in_front_of and behind (R@3 = 0.538 and 0.481 respectively, despite R@5 ≈ 1.000). The model correctly identifies these relations — they appear in the top-5 predictions almost always — but consistently ranks them 4th or 5th rather than 1st through 3rd. The root cause is depth ambiguity. In 3RScan room-scale scenes, objects are distributed across several metres of depth, producing large normalised delta_y values that the model associates with high confidence for in_front_of and behind. On tabletop scenes, objects are densely packed within 0.3–0.8 metres, and the Y separations between objects — after normalisation by avg_diag — are much smaller. The model interprets small normalised delta_y as low confidence for in_front_of, ranking directional alternatives higher instead. This is not a failure of the model to learn the relation; it is a miscalibration of confidence at the boundary of the learned distribution.

Lower_than (R@5 = 0.729) is the only relation where R@5 itself is substantially below 1.000, making it the true failure case. On tabletop scenes, all objects sit on the same flat

surface within a narrow vertical band of a few centimetres. The Z differences between their centroids are smaller, in normalised terms, than anything the model encountered during training on 3RScan — where objects on different shelves, floors, or surfaces are separated by tens of centimetres to metres. The z_rank feature partially compensates for this, but even ordinal rank becomes unreliable when Z differences are within measurement noise. The model genuinely cannot distinguish lower_than from higher_than for objects at nearly identical heights, which is a principled failure: the information required to make the distinction confidently is not present in the scene.

These findings suggest a clear path for improving cross-domain robustness: per-relation threshold re-calibration on a small number of target-domain scenes, combined with additional scale-adaptive normalisation for contact relation features. Even a handful of labelled pairs from the target domain would be sufficient to shift the on_top_of and lower_than thresholds to the tabletop operating point, potentially recovering the remaining R@3 deficit without retraining the base model.

5.6 Scale Domain Gap as Root Cause of Tabletop Degradation

The cross-domain evaluation encompasses two distinct test settings: the LERF scenes and the custom tabletop scenes. The contrast between their results is striking and highly informative. On the LERF scenes — indoor environments at 1–5 metre scale — GeoKAN-Gamma achieves mR@5 = 0.929 with no adaptation. On the tabletop scenes — environments at 0.3–0.8 metre scale — the model achieves R@5 = 0.927 overall but with substantially weaker R@3 = 0.758 and specific relation failures on lower_than and depth-axis relations. The LERF scenes represent a different dataset, different object categories, different lighting conditions, and different reconstruction quality compared to 3RScan — yet performance is strong. The tabletop scenes differ from 3RScan primarily in scale, yet performance degrades measurably on specific relations. This comparison isolates scale as the root cause of the observed degradation.

The mechanism operates through the normalised feature distributions. All node and edge features are normalised by scene-level statistics — specifically avg_diag, the mean object bounding box diagonal. In a 3RScan room, avg_diag is typically 0.8–1.2 metres; on a tabletop, avg_diag is typically 0.10–0.20 metres. This normalisation makes the features dimensionless and theoretically scale-invariant, but the normalisation captures only the first-order scale difference. Second-order effects remain: the distribution of normalised vert_gap values for objects on a tabletop is not identical to the distribution of normalised vert_gap values for objects in a room, because the relative geometry of tabletop objects — all resting on a single surface at nearly identical heights — differs structurally from room objects that may be on different shelves, floors, or wall mountings. The GeoKAN metric, calibrated on the 3RScan distribution of normalised features, is subtly miscalibrated for the tabletop distribution even after normalisation.

A diagnostic comparison confirms this interpretation. The LERF scenes, whilst from a completely different dataset, have a scale of 1–5 metres that is within the same order of magnitude as the 3RScan training distribution of 3–8 metres. The normalised feature

distributions on LERF are therefore close to the training distribution, and the calibrated thresholds transfer without significant degradation. The tabletop scenes, at 0.3–0.8 metres, are one order of magnitude smaller than the training distribution. Whilst the normalisation closes most of this gap, the residual distributional shift is sufficient to affect the relations — lower_than, in_front_of, behind — whose geometric signals are smallest in absolute normalised terms on the tabletop.

This analysis has a practical implication that goes beyond this project. It suggests that the scale-invariance of geometry-first scene graph models is approximate rather than exact: scale-invariant normalisation eliminates gross scale differences but does not eliminate distributional shifts that arise from structural differences in how objects at different scales relate to one another. A model trained on room-scale data and applied to micro-scale data (individual objects or scene fragments) or macro-scale data (building interiors or outdoor environments) will exhibit analogous degradation on the relations most sensitive to the scale gap. Achieving true scale invariance would likely require either training data spanning a wider range of scales or an explicit scale-adaptive normalisation mechanism that adjusts the feature distribution to the operating scale at inference time.

5.7 Gaussian Shape Features: Potential Explored but Not Exploited

One of the distinguishing properties of the 3D Gaussian Splatting representation — compared to point clouds or mesh reconstructions — is that each Gaussian primitive carries a full 3×3 covariance matrix encoding the local surface geometry at that point. The eigenvalues of this matrix describe the shape of the Gaussian's spatial extent: a large first eigenvalue relative to the second and third indicates an elongated, needle-like Gaussian (characteristic of edges and thin structures); a large first and second eigenvalue relative to the third indicates a flat, pancake-like Gaussian (characteristic of surfaces such as tabletops and walls); near-equal eigenvalues indicate a spherical, isotropic Gaussian (characteristic of compact objects). Aggregating these per-Gaussian shape descriptors over all Gaussians in a cluster produces object-level shape features — planarity, elongation, and sphericity — that are unique to the Gaussian representation and unavailable in point cloud or depth-image-based methods.

These features were computed and stored for every Object3D instance throughout the project (accessible as obj._eigenvalues) but were ultimately excluded from the 10-dimensional node feature vector. Early experiments incorporating planarity, elongation, and sphericity as additional node features produced no consistent improvement in validation Macro F1 and in some configurations caused slight degradation, likely due to noise in the covariance estimates arising from the opacity-weighted averaging procedure used to compute the mean cluster covariance. Gaussian splats trained with NerfStudio splatfacto distribute Gaussians non-uniformly across object surfaces — concentrating them at high-curvature regions and view-dependent highlights — so the mean covariance of a cluster is not a clean descriptor of object shape but a mixture of geometric and photometric effects.

Despite this, the theoretical potential of Gaussian shape features remains compelling. Planarity could provide a direct signal for the surface support detection underlying on_top_of: an object with high planarity on its bottom face resting on an object with high planarity on its top face is a physically consistent stacking configuration that the current geometric rules can only approximate through bounding box proximity. Elongation could improve attached_to detection for wall-mounted objects, which tend to have high elongation along the axis perpendicular to the wall surface. Sphericity could help distinguish compact objects — balls, knobs, handles — from flat objects in configurations where the bounding box geometry alone is ambiguous.

Realising this potential would likely require either a more principled covariance aggregation strategy — for example, weighting by Gaussian opacity and restricting to Gaussians on the object's outer surface — or end-to-end training of the covariance descriptor jointly with the relation prediction objective, so that the representation learns which aspects of the covariance distribution are predictive. This represents a meaningful direction for future work that would leverage what is genuinely distinctive about the Gaussian representation rather than treating it as a decorated point cloud.

5.8 Limitations

Several limitations of the current work should be acknowledged.

Scale domain gap. As established in Section 5.6, the model's cross-domain transfer is approximate rather than exact. The tabletop evaluation demonstrates that relations sensitive to small normalised feature values — lower_than, in_front_of, behind — degrade when the operating scale is one order of magnitude below the training scale. The current normalisation strategy mitigates but does not eliminate this distributional shift. A model intended for deployment across a wide range of scene scales would require either multi-scale training data or an explicit scale-adaptive normalisation mechanism.

Static scenes only. The pipeline assumes a static scene during video capture and reconstruction. Objects that move between frames produce reconstruction artefacts — blurred or split Gaussian clusters — that degrade both clustering quality and feature accuracy. Dynamic scenes, scenes with significant human presence, or environments with moving objects are outside the scope of the current system.

Clustering dependency. All downstream relation prediction depends on the quality of the HDBSCAN clustering step. If two physically distinct objects are merged into a single cluster — a failure mode that occurs when objects are in physical contact and share similar colour properties — the resulting node feature vector represents a composite object rather than either individual, and the predicted relations for that cluster will be incorrect. Conversely, if a single object is split into multiple clusters — which occurs with low-saturation or reflective objects — spurious inter-cluster relations will be predicted. The auto-tuning strategy reduces but does not eliminate these failure modes.

Annotation-derived evaluation on 3DSSG. The 3DSSG validation set itself suffers from the annotation sparsity problem that motivates rule-based label injection. Ground truth relations that exist geometrically but were not annotated by human labellers are treated as negatives during evaluation, meaning the reported Macro F1 and R@K metrics understate true performance on the full set of geometrically valid relations. The extent of this underestimation is difficult to quantify precisely without re-annotating the validation set.

Threshold calibration on in-domain data. Per-relation thresholds are calibrated on the 3DSSG validation set, which is drawn from the same distribution as the training data. When these thresholds are applied zero-shot to the tabletop scenes, they are suboptimal for the target domain. The R@K metrics, which are threshold-independent, remain strong cross-domain; the Micro F1 degradation from 0.926 (in-domain) to 0.740 (tabletop, without repair) reflects this threshold mismatch. A small amount of target-domain calibration data — even ten to twenty labelled pairs — would be sufficient to re-calibrate thresholds without retraining.

Closed relation schema. The system predicts a fixed vocabulary of ten spatial relations. Relations outside this schema — such as hanging_from, inside, or touching — cannot be predicted, limiting applicability to tasks that require a richer relation vocabulary. Extending the schema to additional relation types would require both additional training annotations and potentially additional geometric features, as some relations (inside, for example) require volumetric containment reasoning that is only partially captured by the current 22-dimensional edge features.

No object identity at inference. The pipeline predicts spatial relations between anonymous geometric clusters without semantic object labels. Whilst this is consistent with the geometry-first design philosophy and does not affect relation prediction quality, it means the output scene graph contains node identifiers rather than object names unless an optional semantic labelling step (not part of the core pipeline) is applied. Applications that require human-readable scene graphs would need to integrate a separate object recognition component.

6.CONCLUSION

6.1 Summary of Work

This project presented LogicSplat, a geometry-first pipeline for spatial scene graph generation from 3D Gaussian Splats. The system was designed around a single guiding principle: that spatial relations between objects in a three-dimensional scene are fundamentally geometric predicates, and that a model trained on well-engineered geometric features with accurate supervision should be able to predict them accurately without any foundation model, depth sensor, or semantic label at inference time.

The pipeline proceeds from a short smartphone video through COLMAP structure-from-motion and NerfStudio splatfacto reconstruction to produce a Gaussian splat, which is then cleaned via opacity filtering and statistical outlier removal, segmented into object instances via HDBSCAN clustering with Otsu auto-tuning, and represented as a graph with 10-dimensional scale-invariant node features and 22-dimensional edge features

encoding directional offsets, contact geometry, and spatial relationships. Rule-based label injection addresses the annotation sparsity inherent in the 3DSSG training data, converting approximately 90 per cent of false negative training pairs into accurate pseudo-labels and increasing the effective training signal by 8–11 times. GeoKANRelationGNN — a dual-head graph neural network with GATv2Conv message passing and GeoKAN classification heads — predicts ten spatial relation classes per directed object pair. Asymmetric loss, per-relation loss weighting, label smoothing, data augmentation, and per-relation threshold calibration constitute the training strategy. SceneGraphRepair, a zero-parameter symbolic consistency module, enforces inverse completeness, mutual exclusion, asymmetry, and transitivity constraints on the predicted graph through fixed-point iteration.

The system was evaluated on three distinct settings. On the 3DSSG validation benchmark of 85 held-out 3RScan scenes, GeoKAN-Gamma achieved Macro F1 = 0.926, R@3 = 0.878, and R@5 = 0.983 — surpassing all compared methods including the state-of-the-art ReLaGS (R@3 = 0.790, R@5 = 0.870) whilst using 0.07 per cent of its parameter count and completing inference in under two seconds. On the four LERF held-out scenes, the model achieved mR@5 = 0.929 after symbolic repair, exceeding GaussianGraph's best result of 0.632 by 29.7 percentage points. On eight custom-captured tabletop scenes evaluated zero-shot across a substantial scale domain gap, the model achieved R@3 = 0.758 and R@5 = 0.927, with symbolic repair providing an additional 3.4 Micro F1 points via inverse completion with no retraining.

6.2 Novel Contributions

This project makes the following novel contributions to the field of 3D scene graph generation:

C1. Geometry-first scene graph generation from Gaussian Splats without foundation models. This project is the first to demonstrate that a fixed-vocabulary spatial scene graph can be predicted from 3D Gaussian Splat geometry alone — without any vision-language model, segmentation model, or semantic label at inference time — and that this approach outperforms all current foundation-model-dependent methods on the 3DSSG benchmark. The result establishes a new performance ceiling for the task and challenges the assumption that semantic grounding is necessary for geometric spatial relation prediction.

C2. Rule-based label injection for annotation-sparse 3D scene graph datasets. The systematic derivation and injection of geometric pseudo-labels for unannotated directed pairs in 3DSSG — with confidence values calibrated to the certainty of each rule type — is a novel treatment of the annotation sparsity problem in 3D scene graph training. The injection increases the effective training signal by 8–11 times and is the primary driver of the performance jump from Macro F1 = 0.505 to 0.926. Unlike data augmentation, it does not introduce synthetic variation; it corrects false negative labels using geometric facts that are provably correct for directional relations.

C3. Application of GeoKAN to structured spatial prediction. This project is the first application of the GeoKAN architecture — Geometric Kolmogorov-Arnold Networks with a learnable diagonal Riemannian metric — to scene graph generation. The ablation study comparing Gamma, RBF, and Wavelet variants under identical conditions provides the first systematic evaluation of GeoKAN basis function choices on a real structured prediction task, demonstrating that the fixed diagonal metric variant is superior to adaptive variants for well-engineered geometric feature inputs.

C4. Zero-parameter symbolic consistency repair for scene graphs. The SceneGraphRepair module provides a formal, domain-agnostic mechanism for enforcing logical consistency constraints on predicted scene graphs through fixed-point iteration. Its application across three evaluation settings demonstrates that the primary value of the module lies in inverse completion rather than contradiction removal — a finding that reveals a systematic asymmetry in the model's cross-domain predictions and explains why symbolic repair provides disproportionately large gains on on_top_of (+0.298 F1) and lower_than (+0.152 F1) in the cross-domain setting.

C5. Honest cross-domain evaluation methodology. The project provides a documented account of the 11-percentage-point difference between an inflated R@3 = 0.864, obtained through post-hoc ground truth adjustment motivated by model failures, and the honest zero-shot result of R@3 = 0.758. The final evaluation uses ground truth constructed entirely from deterministic geometric principles, with all GT-engineering attempts documented and reverted. This contribution is methodological rather than technical, but it directly addresses a reproducibility concern in cross-domain evaluation that is relevant to the broader scene understanding community.

C6. Scale domain gap characterisation for geometry-first models. The comparative analysis of LERF (in-domain scale, strong performance) and tabletop (out-of-domain scale, degraded performance on specific relations) isolates scale as the primary determinant of cross-domain transfer for geometry-first scene graph models, and identifies the specific relations — lower_than, in_front_of, behind — whose predictions are most sensitive to the scale distribution shift. This characterisation provides a principled basis for future work on scale-adaptive normalisation.

6.3 Future Work

Several directions emerge naturally from the findings and limitations of this project.

Gaussian covariance features for contact relation prediction. As discussed in Section 5.7, the per-Gaussian covariance matrices encode local surface geometry — planarity, elongation, and sphericity — that is unique to the 3DGS representation and currently unused in the node feature vector. Future work should investigate more principled covariance aggregation strategies, such as opacity-weighted surface-restricted averaging or end-to-end learning of covariance descriptors jointly with the relation prediction objective. Planarity in particular is a strong candidate for improving on_top_of and

under prediction, as it directly encodes the surface flatness required for physical support detection.

Scale-adaptive normalisation for wider domain transfer. The current normalisation strategy — dividing by avg_diag — closes the first-order scale gap but leaves residual distributional shifts for relations involving small absolute feature values at tabletop scale. A scale-adaptive mechanism that explicitly identifies the operating scale regime — for example, by estimating the distribution of normalised feature values and shifting the calibrated thresholds accordingly — could extend robust zero-shot transfer to micro-scale environments such as table-top manipulation tasks and macro-scale environments such as building interiors or outdoor scenes.

Few-shot threshold re-calibration for cross-domain deployment. The tabletop results demonstrate that the primary source of Micro F1 degradation cross-domain is threshold miscalibration rather than ranking failure (R@5 remains strong at 0.927). As few as ten to twenty labelled object pairs from a target-domain scene would be sufficient to re-calibrate the ten per-relation thresholds without retraining the base model. A practical deployment protocol that includes a minimal annotation step for threshold adaptation would substantially close the remaining cross-domain gap at negligible annotation cost.

Extended relation schema. The current ten-relation schema excludes physically meaningful relations such as inside, hanging_from, and touching. Extending the schema would require additional positive training examples — the primary reason for their exclusion was insufficient positive rate in 3DSSG — and potentially additional edge features. The 3D containment features already present in the 22-dimensional edge vector (features 17–21) partially address the inside relation; with sufficient training signal, these features may support its addition without major architectural changes.

Dynamic scene handling. The current pipeline assumes static scenes during video capture. Extending it to handle moving objects would require either multi-frame Gaussian tracking — identifying Gaussian primitives that change position between frames — or integration with a separate dynamic object segmentation step prior to relation prediction. This extension would significantly broaden the applicability of the pipeline to robotic manipulation, activity monitoring, and augmented reality settings where scene dynamics are the norm rather than the exception.

Semantic labelling integration. The current system produces scene graphs with anonymous geometric cluster nodes. Integrating a lightweight object recognition step — for example, projecting cluster centroids into video frames and applying a small object classifier — would produce human-readable scene graphs without compromising the geometry-first design of the relation prediction stage. The optional YOLO-based labelling mechanism already present in the system provides a starting point for this integration.

Broader benchmark evaluation. This project evaluates on 3DSSG, LERF, and custom tabletop scenes. Evaluation on additional benchmarks — including 3RScan's RIO10 re-localisation test scenes, the ScanNet200 benchmark, or purpose-built scene graph

datasets for robotics — would provide a more comprehensive characterisation of the system's generalisation properties and facilitate fairer comparison with methods that report results on those benchmarks.

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